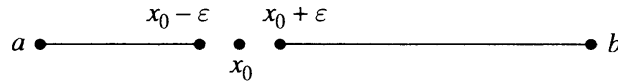


Suppose that $f(x)$ is continuous for $a \leq x < x_0$ and for $x_0 < x \leq b$. We define the **principal value** of the integral $\int_a^b f(x) dx$ to be

$$(6.1) \quad \text{PV} \int_a^b f(x) dx = \lim_{\varepsilon \rightarrow 0} \left(\int_a^{x_0 - \varepsilon} + \int_{x_0 + \varepsilon}^b \right) f(x) dx,$$

provided that the limit exists. The principal value of the integral coincides with the usual value of the (proper or improper) integral if $f(x)$ is absolutely integrable. The same definition is used when the endpoints are infinite or when $f(x)$ is not continuous at the endpoints, provided that $f(x)$ is absolutely integrable over each of the intervals $(a, x_0 - \varepsilon)$ and $(x_0 + \varepsilon, b)$. If $f(x)$ has a finite number of discontinuities within (a, b) , we define the principal value of the integral by dividing the interval (a, b) into subintervals, each containing one discontinuity of $f(x)$, and adding the principal values of the integrals corresponding to the subintervals.



Example. We illustrate by deriving the identity

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x^3 - 1} dx = -\frac{\pi}{\sqrt{3}},$$

Observe first that the integrand behaves like $1/(x - 1)$ near $x = 1$, so that the integral is absolutely divergent. It is absolutely convergent on each interval $(-\infty, 1 - \varepsilon)$ and $(1 + \varepsilon, \infty)$, and the principal value is defined by

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x^3 - 1} dx = \lim_{\varepsilon \rightarrow 0} \left(\int_{-\infty}^{1 - \varepsilon} + \int_{1 + \varepsilon}^{\infty} \right) \frac{1}{x^3 - 1} dx.$$

We consider the function $f(z) = 1/(z^3 - 1)$, which has simple poles at the cube roots of unity. For $\varepsilon > 0$ small and $R > 0$ large, we consider the indented half-disk D in the upper half-plane, consisting of z in the upper half-plane satisfying $|z| < R$ and $|z - 1| > \varepsilon$. The function $f(z)$ has one pole in D , a simple pole at $z = e^{2\pi i/3}$. For the residue, Rule 4 gives

$$\text{Res} \left[\frac{1}{z^3 - 1}, e^{2\pi i/3} \right] = \frac{1}{3z^2} \Big|_{z=e^{2\pi i/3}} = \frac{e^{2\pi i/3}}{3}.$$

The residue theorem yields

$$(6.2) \quad \int_{\partial D} f(z) dz = 2\pi i \frac{e^{2\pi i/3}}{3} = -\frac{\pi}{\sqrt{3}} - \frac{\pi}{3}i.$$

The integral around ∂D breaks into the sum of four integrals, from $-R$ to $1 - \varepsilon$, around a semicircular contour C_ε centered at 1 of radius ε in the clockwise direction, from $1 + \varepsilon$ to R , and around the semicircular contour Γ_R

of radius R in the counterclockwise direction,

$$(6.3) \quad \int_{\partial D} f(z) dz = \left(\int_{-R}^{1-\varepsilon} + \int_{C_\varepsilon} + \int_{1+\varepsilon}^R + \int_{\Gamma_R} \right) f(z) dz.$$

The ML -estimate gives

$$\left| \int_{\Gamma_R} \frac{1}{z^3 - 1} dz \right| \leq \frac{1}{R^3 - 1} \cdot \pi R \sim \frac{1}{R^2},$$

which tends to 0 as $R \rightarrow \infty$. We combine (6.2) and (6.3), and we pass to the limit as $R \rightarrow \infty$, to obtain

$$\left(\int_{-\infty}^{1-\varepsilon} + \int_{1+\varepsilon}^{\infty} \right) \frac{1}{x^3 - 1} dx + \int_{C_\varepsilon} \frac{1}{z^3 - 1} dz = -\frac{\pi}{\sqrt{3}} - \frac{\pi}{3}i.$$

From Rule 4, the residue of $f(z)$ at $z = 1$ is

$$\text{Res} \left[\frac{1}{z^3 - 1}, 1 \right] = \frac{1}{3z^2} \Big|_{z=1} = \frac{1}{3}.$$

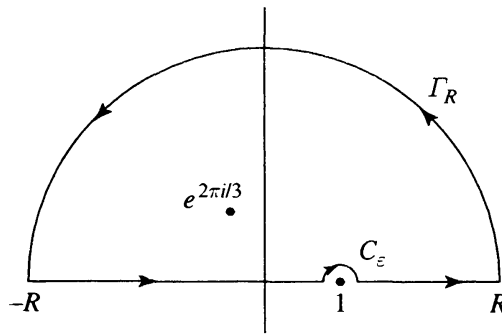
The fractional residue theorem, with angle $-\pi$, then yields

$$\lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \frac{1}{z^3 - 1} dz = -\frac{\pi}{3}i.$$

If we pass to the limit in (6.3) as $\varepsilon \rightarrow 0$, we then obtain

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x^3 - 1} dx - \frac{\pi}{3}i = -\frac{\pi}{\sqrt{3}} - \frac{\pi}{3}i.$$

If we add $\pi i/3$ to both sides, we obtain our asserted identity.



Principal value integrals are important in harmonic analysis. One of the most important integral operators in analysis is the singular integral operator called the **Hilbert transform**, defined by

$$(Hu)(t) = \text{PV} \int_{-\infty}^{\infty} \frac{u(s)}{s - t} ds, \quad -\infty < t < \infty,$$

where $u(s)$ is an integrable function on the real line. It turns out that the principal value exists for “almost all” real numbers t and defines a “good”

operator $u \mapsto Hu$. If $u(s)$ is extended appropriately to be harmonic in the upper half-plane, then $(Hu)(s)$ is the boundary value function of its harmonic conjugate.

Exercises for VII.6

1. Integrate $1/(1-x^2)$ directly, using partial fractions, and show that

$$\text{PV} \int_0^\infty \frac{dx}{1-x^2} = 0.$$

Show that

$$\int_0^1 \frac{dx}{1-x^2} = +\infty, \quad \int_1^\infty \frac{dx}{1-x^2} = -\infty.$$

2. Obtain the principal value in Exercise 1 by taking imaginary parts of the identity (5.4) in the preceding section and making a change of variable.
3. By integrating around the boundary of an indented half-disk in the upper half-plane, show that

$$\text{PV} \int_{-\infty}^\infty \frac{1}{(x^2+1)(x-a)} dx = -\frac{\pi a}{a^2+1}, \quad -\infty < a < \infty.$$

4. Suppose $m \geq 2$ and $a_1 < a_2 < \cdots < a_m$. By integrating around the boundary of an indented half-disk in the upper half-plane, show that

$$\text{PV} \int_{-\infty}^\infty \frac{1}{(x-a_1)(x-a_2)\cdots(x-a_m)} dx = 0.$$

5. Show that

$$\text{PV} \int_0^\infty \frac{x^{a-1}}{x^b-1} dx = -\frac{\pi}{b} \cot\left(\frac{\pi a}{b}\right), \quad 0 < a < b.$$

Hint. For $b > 1$ one can integrate a branch of $z^{a-1}/(z^b-1)$ around a sector of aperture $2\pi/b$, indented at $z=1$ and at $z=e^{2\pi i/b}$.

6. By integrating a branch of $(\log z)/(z^b-1)$ around an indented sector of aperture $2\pi/b$, show that for $b > 1$,

$$\int_0^\infty \frac{\log x}{x^b-1} dx = \frac{\pi^2}{b^2 \sin^2(\pi/b)}, \quad \text{PV} \int_0^\infty \frac{1}{x^b-1} dx = -\frac{\pi}{b} \cot(\pi/b)$$

7. Suppose that $P(z)$ and $Q(z)$ are polynomials, $\deg Q(z) \geq \deg P(z) + 2$, and the zeros of $Q(z)$ on the real axis are all simple. Show that

$$\text{PV} \int_{-\infty}^\infty \frac{P(x)}{Q(x)} dx = 2\pi i \sum \text{Res} \left[\frac{P(z)}{Q(z)}, z_j \right] + \pi i \sum \text{Res} \left[\frac{P(z)}{Q(z)}, x_k \right]$$

summed over the poles z_j of $P(z)/Q(z)$ in the open upper half-plane and the poles x_k of $P(z)/Q(z)$ on the real axis. *Remark.* In other words, the principal value of the integral is $2\pi i$ times the sum of the residues in the upper half-plane, where we count the poles on the real axis as being half in and half out of the upper half-plane.

7. Jordan's Lemma

Another class of absolutely divergent integrals is made up of integrals of the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \sin x \, dx, \quad \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} \cos x \, dx,$$

where $P(z)$ and $Q(z)$ are polynomials satisfying

$$\deg Q(z) = \deg P(z) + 1.$$

To evaluate such an integral we would replace the sine or cosine function by e^{iz} and integrate over the boundary of the usual half-disk in the upper half-plane. If $\deg Q(z) \geq \deg P(z) + 2$, then the *ML*-estimate shows that the integral over the semicircular piece Γ_R of the boundary tends to 0 as $R \rightarrow \infty$. If $\deg Q(z) = \deg P(z) + 1$, the integral over Γ_R still tends to 0, but this is not so obvious. It is a consequence of the following estimate.

Lemma (Jordan's Lemma). *If Γ_R is the semicircular contour $z(\theta) = Re^{i\theta}$, $0 \leq \theta \leq \pi$, in the upper half-plane, then*

$$(7.1) \quad \int_{\Gamma_R} |e^{iz}| |dz| < \pi.$$

For the parameterization $z(\theta) = Re^{i\theta}$ we have $|e^{iz}| = e^{R \sin \theta}$ and $|dz| = R \, d\theta$, so the estimate (7.1) becomes

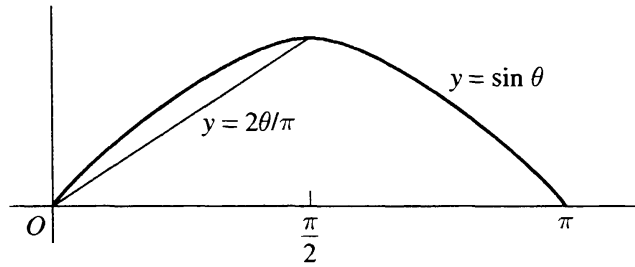
$$(7.2) \quad \int_0^\pi e^{-R \sin \theta} d\theta < \frac{\pi}{R}.$$

In order to establish (7.2), note that $\sin \theta$ is concave down on the interval $0 \leq \theta \leq \pi/2$, so the graph of $\sin \theta$ lies above the straight line connecting its endpoints,

$$\sin \theta \geq 2\theta/\pi, \quad 0 \leq \theta \leq \pi/2.$$

Thus

$$\begin{aligned} \int_0^\pi e^{-R \sin \theta} d\theta &= 2 \int_0^{\pi/2} e^{-R \sin \theta} d\theta \leq 2 \int_0^{\pi/2} e^{-2R\theta/\pi} d\theta \\ &= \frac{\pi}{R} \int_0^{1/R} e^{-t} dt < \frac{\pi}{R} \int_0^\infty e^{-t} dt = \frac{\pi}{R}. \end{aligned}$$



Example. As an application of Jordan's lemma, we show that

$$(7.3) \quad \lim_{R \rightarrow \infty} \int_0^R \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

Note that $(\sin z)/z$ is analytic at $z = 0$, so that $(\sin x)/x$ is absolutely integrable over any finite interval. It is not difficult to see (Exercise 1) that

$$\lim_{R \rightarrow \infty} \int_0^R \frac{|\sin x|}{x} dx = +\infty.$$

Thus the integral over $[0, \infty)$ is absolutely divergent. Since the integrand is an even function, (7.3) is equivalent to

$$(7.4) \quad \lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} dx = \pi.$$

We may think of the limit (7.4) as a principal value at infinity,

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{\sin x}{x} dx = \text{PV} \int_{-\infty}^{\infty} \frac{\sin x}{x} dx.$$

To evaluate the limit, we consider the function $f(z) = e^{iz}/z$, which has only one pole, a simple pole at $z = 1$ with residue 1. Let D be the indented half-disk consisting of points z in the upper half-plane satisfying $\varepsilon < |z| < R$. The integral around ∂D breaks into the sum of four integrals, from $-R$ to $-\varepsilon$, around a semicircular contour C_ε centered at 0 of radius ε in the clockwise direction, from ε to R , and around the semicircular contour Γ_R of radius R in the counterclockwise direction. Since $f(z)$ is analytic on D , we may apply Cauchy's theorem, to obtain

$$(7.5) \quad 0 = \int_{\partial D} f(z) dz = \left(\int_{-R}^{-\varepsilon} + \int_{C_\varepsilon} + \int_{\varepsilon}^R + \int_{\Gamma_R} \right) f(z) dz.$$

The integral over C_ε is handled by the fractional residue formula, with angle $-\pi$,

$$\lim_{\varepsilon \rightarrow 0} \int_{C_\varepsilon} \frac{e^{iz}}{z} dz = -\pi i \operatorname{Res} \left[\frac{e^{iz}}{z}, 0 \right] = -\pi i.$$

Passing to the limit in (7.5) as $\varepsilon \rightarrow 0$, we obtain

$$0 = \text{PV} \int_{-R}^R \frac{e^{ix}}{x} dx - \pi i + \int_{\Gamma_R} \frac{e^{iz}}{z} dz.$$

Taking the imaginary part of this identity, we obtain

$$\int_{-R}^R \frac{\sin x}{x} dx + \operatorname{Im} \int_{\Gamma_R} \frac{e^{iz}}{z} dz = \pi.$$

Now the integral over Γ_R is handled by Jordan's lemma,

$$\left| \int_{\Gamma_R} \frac{e^{iz}}{z} dz \right| \leq \frac{1}{R} \int_{\Gamma_R} |e^{iz}| |dz| < \frac{\pi}{R}.$$

Since this tends to 0 as $R \rightarrow \infty$, in the limit we obtain (7.4).

Exercises for VII.7

1. Show that

$$\int_0^\infty \frac{|\sin x|}{x} dx = +\infty.$$

Hint: Show that the area under the m th arch of $|\sin x|/x$ is $\sim 1/m$.

2. Show that

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{x^3 \sin x}{(x^2 + 1)^2} dx = \frac{\pi}{2e}.$$

3. Evaluate the limits

$$\lim_{R \rightarrow \infty} \int_{-R}^R \frac{x \sin(ax)}{x^2 + 1} dx, \quad -\infty < a < +\infty.$$

Show that they do not depend continuously on the parameter a .

4. By integrating $z^{a-1}e^{iz}$ around the boundary of a domain in the first quadrant bounded by the real and imaginary axes and a quarter-circle, show that

$$\lim_{R \rightarrow \infty} \int_0^R x^{a-1} \cos x dx = \Gamma(a) \cos(\pi a/2), \quad 0 < a < 1,$$

$$\lim_{R \rightarrow \infty} \int_0^R x^{a-1} \sin x dx = \Gamma(a) \sin(\pi a/2), \quad 0 < a < 1,$$

where $\Gamma(a)$ is the gamma function defined by

$$\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt.$$

Remark: The formula for the sine integral holds also for $-1 < a < 0$. To see this, integrate by parts.

5. Show that

$$\lim_{R \rightarrow \infty} \int_0^R \sin(x^2) dx = \lim_{R \rightarrow \infty} \int_0^R \cos(x^2) dx = \frac{\sqrt{\pi}}{2\sqrt{2}},$$

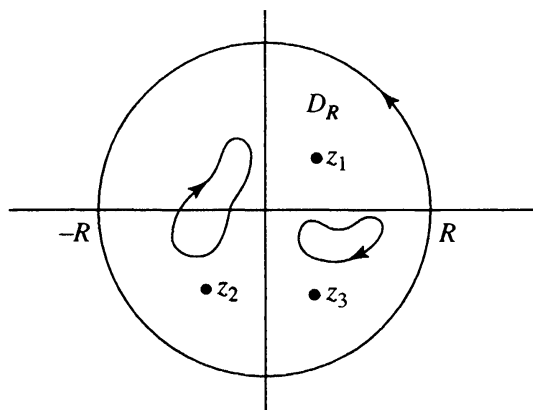
by integrating e^{iz^2} around the boundary of the pie-slice domain determined by $0 < \arg z < \pi/4$ and $|z| < R$. *Remark.* These improper integrals are called the **Fresnel integrals**. The identities can also be deduced from the preceding exercise by changing variable.

8. Exterior Domains

An **exterior domain** is a domain D in the complex plane that includes all large z , that is, D includes all z such that $|z| \geq R$ for some R . The residue theorem is valid also for exterior domains, though the residue formula must take into account the point at ∞ .

Theorem. *Let D be an exterior domain with piecewise smooth boundary. Suppose that $f(z)$ is analytic on $D \cup \partial D$, except for a finite number of isolated singularities $z_1, \dots, z_m$ in D , and let a_{-1} be the coefficient of $1/z$ in the Laurent expansion $f(z) = \sum a_k z^k$ that converges for $|z| > R$. Then*

$$(8.1) \quad \int_{\partial D} f(z) dz = -2\pi i a_{-1} + 2\pi i \sum_{j=1}^m \operatorname{Res}[f(z), z_j].$$



Here the boundary ∂D is oriented as usual, so that D lies on its left as ∂D is traversed in the positive direction. The formula is proved by applying the usual residue theorem to the bounded domain D_R consisting of $z \in D$ such that $|z| < R$. This yields

$$\int_{\partial D} f(z) dz + \int_{|z|=R} f(z) dz = \int_{\partial D_R} f(z) dz = 2\pi i \sum_{j=1}^m \operatorname{Res}[f(z), z_j].$$

If we substitute the Laurent series expansion for $f(z)$ into the integral over the circle $\{|z| = R\}$ and integrate term by term, we obtain $2\pi i a_{-1}$ for the integral. This is simply the integral formula for the Laurent coefficient a_{-1} . In any event, we move this summand to the right-hand side, picking up a minus sign, and we obtain (8.1).

Example. We show using contour integration that

$$\int_0^1 \frac{1}{\sqrt{x(1-x)}} dx = \pi.$$

Consider the function $1/\sqrt{z(1-z)}$, which has two analytic branches on the slit plane $\mathbb{C} \setminus [0, 1]$. The branches are analytic at ∞ , and the Laurent expansions are obtained in terms of a binomial series:

$$\begin{aligned} \frac{1}{\sqrt{z(1-z)}} &= \pm \frac{i}{z} \left(1 - \frac{1}{z}\right)^{-1/2} \\ &= \pm \frac{i}{z} \left[1 + (-1/2)(-1/z) + \frac{(-1/2)(-3/2)}{2!}(-1/z)^2 + \cdots\right] \\ &= \pm i \left[\frac{1}{z} + \frac{1}{2z^2} + \frac{3}{8z^3} + \cdots\right], \quad |z| > 1. \end{aligned}$$

Let $f(z)$ be the branch of the function that is positive on the top edge of the slit $[0, 1]$. Since the phase factor of $f(z)$ is -1 at each of the branch points 0 and 1 , $f(z)$ is negative on the bottom edge of the slit. By following the argument of $1/\sqrt{1-z}$ around a small semicircle from the top edge of $[0, 1]$ to the interval $(1, +\infty)$, which increases by $\pi/2$, we see that the values $f(x)$ are positive imaginary for $x > 1$. Consequently, our branch $f(z)$ corresponds to the $+$ sign in the Laurent expansion, and the residue at ∞ is given by

$$\text{Res}[f(z), \infty] = -a_{-1} = -i.$$

We integrate $f(z)$ around the **dogbone contour** Γ_ε (or **dumbbell contour**), traversing the top edge of the slit from ε to $1-\varepsilon$, a circle γ_ε centered at 1 of radius ε , the bottom edge of the slit from $1-\varepsilon$ to ε , and a circle C_ε centered at 0 of radius ε . If we apply the residue formula (8.1) to the exterior domain lying outside the dogbone contour, we obtain

$$(8.2) \quad \int_{\Gamma_\varepsilon} f(z) dz = 2\pi i \text{Res}[f(z), \infty] = 2\pi i(-i) = 2\pi.$$

The integrals over the circles γ_ε and C_ε are bounded by the *ML*-estimate

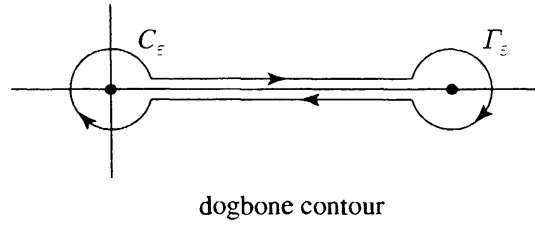
$$\begin{aligned} \left| \int_{C_\varepsilon} f(z) dz \right| &\leq \frac{2}{\sqrt{\varepsilon}} \cdot 2\pi\varepsilon \sim \sqrt{\varepsilon}, \\ \left| \int_{\gamma_\varepsilon} f(z) dz \right| &\leq \frac{2}{\sqrt{\varepsilon}} \cdot 2\pi\varepsilon \sim \sqrt{\varepsilon}. \end{aligned}$$

Since the values of $f(z)$ on the bottom edge of the slit are minus those on the top edge, while the direction is reversed, the integral along the bottom edge is equal to the integral along the top edge. Thus we obtain



$$\int_{\Gamma_\varepsilon} f(z) dz \rightarrow 2 \int_0^1 \frac{1}{\sqrt{x(1-x)}} dx$$

as $\varepsilon \rightarrow 0$. If we combine this with (8.2), we obtain the desired identity.



Note that it was not necessary to determine the choice of sign in the Laurent expansion of $f(z)$. If we had carried along the $\pm$ sign, we would have arrived at a value for the integral of $\pm\pi$. Since the integrand is positive, the value $+\pi$ must be the correct choice. Working backwards, we see that our choice of the $+$ sign in the Laurent expansion of $f(z)$ was correct.

Suppose now that $f(z)$ is analytic for $|z| \geq R$, with Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n, \quad |z| \geq R.$$

We define the **residue of $f(z)$ at ∞** to be

$$(8.3) \quad \text{Res}[f(z), \infty] = -a_{-1}.$$

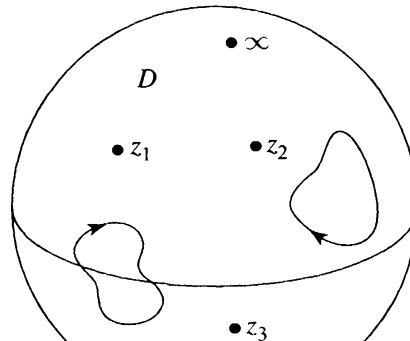
If D_R is the exterior domain $\{|z| > R\}$, this definition is equivalent to

$$\int_{\partial D_R} f(z) dz = 2\pi i \text{Res}[f(z), \infty].$$

The orientation of the circle $\{|z| = R\}$ with respect to D_R is clockwise, and this accounts for the minus sign. With this definition of residue at ∞ , formula (8.1) becomes

$$(8.4) \quad \int_{\partial D} f(z) dz = 2\pi i \text{Res}[f(z), \infty] + 2\pi i \sum_{j=1}^m \text{Res}[f(z), z_j].$$

Thus the residue theorem for exterior domains is the same as for bounded domains, except that we must take the residue at ∞ into account.



Note that the function $f(z)$ can be analytic at ∞ and still have a nonzero residue at ∞ . For instance, the function $1/z$ is analytic at ∞ , while

$$\operatorname{Res} \left[\frac{1}{z}, \infty \right] = -1.$$

There is a mathematical explanation for this state of affairs. The explanation is that in order to have a coordinate-free definition of residues, the residue should not be associated with the function $f(z)$ but with the differential $f(z)dz$. Even though $f(z)$ may be analytic at ∞ , the differential $f(z)dz$ need not be analytic at ∞ , as can be seen by making the usual change of variable $w = 1/z$. (See Exercise 13.)

Exercises for VII.8

1. Evaluate the residue at ∞ of the following functions.

$$\begin{array}{lll} \text{(a)} \frac{z}{z^2 - 1} & \text{(c)} \frac{z^3 + 1}{z^2 - 1} & \text{(e)} z^n e^{1/z}, \quad n = 0, \pm 1, \dots \\ \text{(b)} \frac{1}{(z^2 + 1)^2} & \text{(d)} \frac{z^9 + 1}{z^6 - 1} & \text{(f)} \sqrt{\frac{z - a}{z - b}} \end{array}$$

Note. There are two possibilities for (f), one for each branch of the square root.

2. Show by integrating around the dogbone contour that

$$\int_0^1 \frac{x^4}{\sqrt{x(1-x)}} dx = \frac{35\pi}{128}.$$

3. Fix an integer n , positive or negative. Determine for which complex values of the parameter a the integral

$$\int_0^1 \frac{x^n}{x^a(1-x)^{1-a}} dx$$

converges, and evaluate it.

4. Show that

$$\int_{-1}^1 \frac{\sqrt{1-x^2}}{1+x^2} dx = \pi(\sqrt{2}-1).$$

5. Show that the sum of the residues of a rational function on the extended complex plane is equal to zero.
6. Find the residue of $z/(z^2 + 1)$ at each pole in the extended complex plane, and check that the sum of the residues is zero.
7. Find the sum of the residues of $(3z^4 + 2z + 1)/(8z^5 + 5z^2 + 2)$ at its poles in the (finite) complex plane.

8. Fix $n \geq 1$ and $k \geq 0$. Find the residue of $z^k/(z^n - 1)$ at ∞ by expanding $1/(z^n - 1)$ in a Laurent series. Find the residue of $z^k/(z^n - 1)$ at each finite pole, and verify that the sum of all the residues is zero.

9. Show that if $f(z)$ is analytic at ∞ , then

$$\text{Res}[f(z), \infty] = - \lim_{z \rightarrow \infty} z(f(z) - f(\infty)).$$

10. Let D be an exterior domain. Suppose that $f(z)$ is analytic on $D \cup \partial D$ and at ∞ . Show that

$$\frac{1}{2\pi i} \int_{\partial D} \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) - f(\infty), \quad z \in D.$$

11. If $f(x)$ is not integrable at ∞ , we define the principal value

$$\text{PV} \int_{-\infty}^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \text{PV} \int_{-R}^R f(x) dx.$$

Show that

$$\text{PV} \int_{-\infty}^{\infty} \frac{1}{x - a} dx = \begin{cases} i\pi, & \text{Im } a > 0, \\ 0, & \text{Im } a = 0, \\ -i\pi, & \text{Im } a < 0. \end{cases}$$

12. Suppose that $P(z)$ and $Q(z)$ are polynomials such that the degree of $Q(z)$ is strictly greater than the degree of $P(z)$. Suppose that the zeros $x_1, \dots, x_m$ of $Q(z)$ on the real axis are all simple, and set $x_0 = \infty$. Show that

$$\text{PV} \int_{-\infty}^{\infty} \frac{P(x)}{Q(x)} dx = 2\pi i \sum \text{Res} \left[\frac{P(z)}{Q(z)}, z_j \right] + \pi i \sum \text{Res} \left[\frac{P(z)}{Q(z)}, x_k \right]$$

summed over the poles z_j of $P(z)/Q(z)$ in the open upper half-plane and summed over the x_k 's including ∞ . *Hint.* Use the preceding exercise. See also Exercise 6.7.

13. Show that the analytic differential $f(z)dz$ transforms under the change of variable $w = 1/z$ to $-f(1/w)dw/w^2$. Show that the residue of $f(z)$ at $z = \infty$ coincides with that of $-f(1/w)/w^2$ at $w = 0$.

VIII

The Logarithmic Integral

In this chapter we discuss the argument principle and develop several of its consequences. In Section 1 we derive the argument principle from the residue theorem, and we use the argument principle to locate the zeros of analytic functions. Sections 2 through 5 can be viewed as a study of how the zeros of an analytic function depend on various types of parameters. Sections 6 and 7 are devoted to winding numbers of closed paths and the jump theorem for the Cauchy integral. The jump theorem yields an easy proof of the Jordan curve theorem in the smooth case, and a proof of the full Jordan curve theorem is laid out in the exercises. In Section 8 we introduce simply connected domains and we characterize these in several ways. While the material in this chapter is of fundamental importance for the Riemann mapping theorem in Chapter XI and for various further developments, the student can skip to Chapter IX immediately after Sections 1 and 2.

1. The Argument Principle

Suppose $f(z)$ is analytic on a domain D . For a curve γ in D such that $f(z) \neq 0$ on γ , we refer to

$$(1.1) \quad \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \frac{1}{2\pi i} \int_{\gamma} d \log f(z)$$

as the **logarithmic integral** of $f(z)$ along γ . Thus the logarithmic integral measures the change of $\log f(z)$ along the curve γ . It can be used to count zeros and poles of meromorphic functions. The following theorem is one version of the argument principle.

Theorem. *Let D be a bounded domain with piecewise smooth boundary ∂D , and let $f(z)$ be a meromorphic function on D that extends to be analytic on ∂D , such that $f(z) \neq 0$ on ∂D . Then*

$$(1.2) \quad \frac{1}{2\pi i} \int_{\partial D} \frac{f'(z)}{f(z)} dz = N_0 - N_{\infty},$$



where N_0 is the number of zeros of $f(z)$ in D and N_∞ is the number of poles of $f(z)$ in D , counting multiplicities.

The formula (1.2) is simply the residue theorem for $f'(z)/f(z)$. This function is analytic on $D \cup \partial D$ except possibly at the zeros and poles of $f(z)$ in D . Let z_0 be a zero or pole of $f(z)$, and let N be the order of $f(z)$ at z_0 , that is, N is the order of the zero if z_0 is a zero, and N is minus the order of the pole if z_0 is a pole. Then

$$f(z) = (z - z_0)^N g(z),$$

where $g(z)$ is analytic at z_0 and $g(z_0) \neq 0$. So

$$\frac{f'(z)}{f(z)} = \frac{N(z - z_0)^{N-1}g(z)}{(z - z_0)^N g(z)} + \frac{(z - z_0)^N g'(z)}{(z - z_0)^N g(z)} = \frac{N}{z - z_0} + [\text{analytic}].$$

Thus $f'(z)/f(z)$ has a simple pole at z_0 , with residue N . If now we sum the N 's over the zeros and poles, we find that the sum of the residues of $f'(z)/f(z)$ in D is $N_0 - N_\infty$. Thus (1.2) follows from the residue theorem.

Now we look more carefully at the logarithmic integral (1.1), which we express in the form

$$\frac{1}{2\pi i} \int_\gamma d \log f(z) = \frac{1}{2\pi i} \int_\gamma d \log |f(z)| + \frac{1}{2\pi} \int_\gamma d \arg(f(z)).$$

The differential $d \log |f(z)|$ is exact. If we parametrize the curve γ by $\gamma(t) = x(t) + iy(t)$, $a \leq t \leq b$, then

$$\int_\gamma d \log |f(z)| = \log |f(\gamma(b))| - \log |f(\gamma(a))|,$$

which depends only on the initial point $\gamma(a)$ and the endpoint $\gamma(b)$ of the curve. In particular, $\int_\gamma d \log |f(z)| = 0$ if γ is a closed curve.

The differential $d \arg f(z)$ is closed but not exact. Its integral is computed by choosing a continuous single-valued determination of $\arg f(\gamma(t))$ for $a \leq t \leq b$. Then for this determination,

$$(1.3) \quad \int_\gamma d \arg(f(z)) = \arg f(\gamma(b)) - \arg f(\gamma(a)).$$

This quantity (1.3) is referred to as the **increase in the argument of $f(z)$ along γ** . It is defined for any (continuous) path γ in D . Since any two continuous determinations of $\arg f(\gamma(t))$ differ by a constant, the increase in the argument given by (1.3) is independent of the continuous determination.

If γ is a concatenation of curve segments, the increase in the argument of $f(z)$ along γ is obtained by adding the increases along the various curve segments. If the direction of the curve γ is reversed, the increase in the argument is replaced by its negative. If γ is a closed curve, so that $\gamma(a) = \gamma(b)$, the determinations of $\arg f(\gamma(b))$ and $\arg f(\gamma(a))$ differ by an integral

multiple of 2π , so that the increase in the argument of $f(z)$ around γ is an integral multiple of 2π .

Example. The increase in the argument of $f(z) = (z - z_0)^n$ counterclockwise around the circle $|z - z_0| = \rho$ is determined as follows. We parametrize the circle by $\gamma(t) = z_0 + \rho e^{it}$, $0 \leq t \leq 2\pi$. Then $f(\gamma(t)) = \rho^n e^{int}$, and a continuous determination for the argument of $f(\gamma(t))$ is given by $\arg f(\gamma(t)) = nt$, $0 \leq t \leq 2\pi$. Thus the increase in the argument of $f(z)$ around γ is

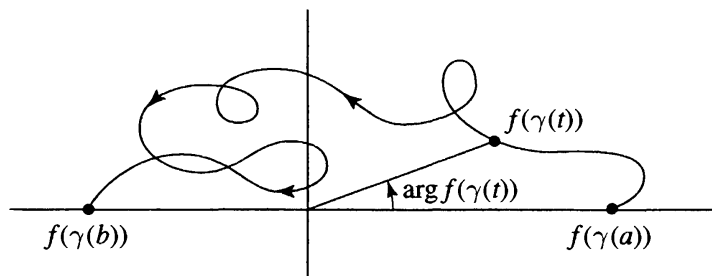
$$\int_{\gamma} d \arg(f(z)) = nt \Big|_{t=0}^{t=2\pi} = 2\pi n.$$

Now we return to a bounded domain D , whose boundary ∂D consists of a finite number of piecewise-smooth closed curves with the usual orientation, so that D lies on the left as we traverse the curves in ∂D in the positive direction. We define the **increase in the argument of $f(z)$ around the boundary of D** to be the sum of its increases around the closed curves in ∂D . The argument principle can be restated as follows.

Theorem. *Let D be a bounded domain with piecewise smooth boundary ∂D , and let $f(z)$ be a meromorphic function on D that extends to be analytic on ∂D , such that $f(z) \neq 0$ on ∂D . Then the increase in the argument of $f(z)$ around the boundary of D is 2π times the number of zeros minus the number of poles of $f(z)$ in D ,*

$$(1.4) \quad \int_{\partial D} d \arg(f(z)) = 2\pi(N_0 - N_{\infty}).$$

Before illustrating the theorem with an example, we note that it is easy to track the argument of $f(z)$ along a segment of a curve in a domain where a single-valued determination of the argument function is available. In this situation, the increase in the argument is obtained simply by evaluating the branch of the argument function at the initial and terminal values of $f(z)$. Suppose, for instance, that the values $f(\gamma(t))$ start from the positive real axis at $t = a$, they lie in the upper half-plane for $a < t < b$, and they hit the negative real axis at $t = b$. Then the increase in the argument of $f(\gamma(t))$ for $a \leq t \leq b$ is π , no matter how wildly the curve might wiggle in the upper half-plane, since it can be evaluated by evaluating a branch of the argument function, in this case the principal branch, at the endpoints. By the same token, if the curve $f(\gamma(t))$ goes from the positive real axis at $t = a$ through the lower half-plane for $a < t < b$ and terminates on the negative real axis, then the increase of the argument of $f(\gamma(t))$ along the curve is $-\pi$, no matter how erratically the curve behaves in the lower half-plane.

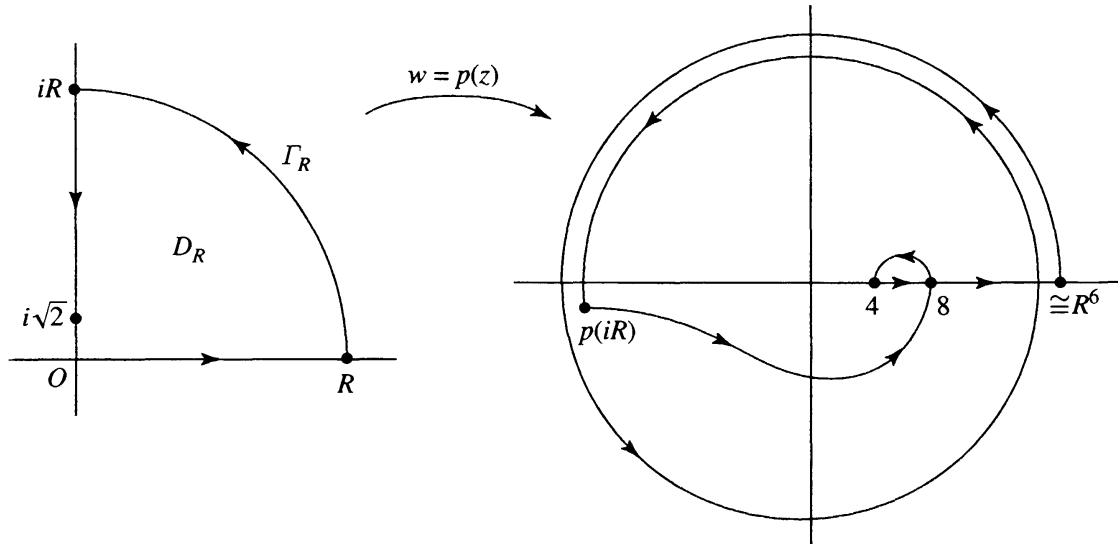


Example. We show how to use the argument principle to find the number of zeros of the polynomial $p(z) = z^6 + 9z^4 + z^3 + 2z + 4$ in the first quadrant $\{0 < \arg z < \pi/2\}$ and in the second quadrant $\{\pi/2 < \arg z < \pi\}$. First note that $p(x) > 0$ for $-\infty < x < \infty$. This is clear if $x \geq 0$, and it follows from the estimates $4 + x^3 + 2x > 0$ for $-1 \leq x \leq 0$ and $9x^4 + x^3 + 2x > 0$ for $x \leq -1$. Since there are no real zeros of $p(z)$, and since the coefficients of $p(z)$ are real, the zeros come in complex conjugate pairs, and there are three zeros in the upper half-plane. To determine the number of zeros in the first quadrant, we estimate the increase of $\arg p(z)$ around the boundary of the quarter disk D_R of a large radius R , consisting of the points z in the first quadrant satisfying $|z| < R$. Since the increase in the argument of $p(z)$ around ∂D_R is 2π times the number of zeros in D_R , any reasonable approximation will yield the exact value. We break ∂D_R into three paths. Along the real axis from 0 to R we have $p(x) > 0$, so that the argument is constant, and the increase is zero. Along the quarter-circle Γ_R defined by $|z| = R$ and $0 \leq \arg z \leq \pi/2$, the term z^6 dominates, and $\arg p(z) \approx 6 \arg z$. The increase in $\arg p(z)$ along Γ_R is thus approximately $6(\pi/2) = 3\pi$. To determine the increase in $\arg p(z)$ down the imaginary axis from iR to 0, we substitute $z = iy$ and we follow the values of

$$p(iy) = -y^6 + 9y^4 + 4 + i(-y^3 + 2y)$$

as y decreases from R to 0. At the initial point of this segment we have $\operatorname{Re} p(iR) \approx -R^6$ and $\operatorname{Im} p(iR) \approx -R^3$, so that $p(iR)$ lies in the third quadrant and has argument approximately $-\pi$. At the terminal point we have $p(0) = 4$ on the positive real axis. To see how $\arg p(iy)$ behaves, we determine where the curve $p(iy)$ crosses the x -axis. The curve crosses the real axis for parameter values y that satisfy $-y^3 + 2y = 0$. The solutions of this cubic are $y = 0$ and $y = \pm\sqrt{2}$. There is only one crossing point corresponding to $y > 0$, at $y = \sqrt{2}$. The crossing point is at $p(i\sqrt{2}) = 32$, which lies on the positive real axis. Thus $p(iy)$ remains in the lower half-plane as the parameter y decreases from $y = R$ to $y = \sqrt{2}$, and it hits the positive real axis at $y = \sqrt{2}$. Thus we see that the increase in the argument of $p(iy)$ as y decreases from R to $\sqrt{2}$ is approximately π . Since $p(iy)$ is in the upper half-plane for y between $\sqrt{2}$ and 0, and since the starting and ending points for this segment of the curve are on the positive real axis, the increase in the argument of $p(iy)$ as y decreases from $\sqrt{2}$ to 0 is zero. Thus the total increase of $\arg p(z)$ around ∂D_R is approximately $3\pi + \pi = 4\pi$,

hence it is exactly 4π . By the argument principle, $p(z)$ has two zeros in the first quadrant. We have seen that there are no zeros on the imaginary axis, so the remaining zero of $p(z)$ in the upper half-plane lies in the second quadrant.



Exercises for VIII.1

1. Show that $z^4 + 2z^2 - z + 1$ has exactly one root in each quadrant.
2. Find the number of zeros of the polynomial $p(z) = z^4 + z^3 + 4z^2 + 3z + 2$ in each quadrant.
3. Find the number of zeros of the polynomial $p(z) = z^6 + 4z^4 + z^3 + 2z^2 + z + 5$ in the first quadrant $\{\operatorname{Re} z > 0, \operatorname{Im} z > 0\}$.
4. Find the number of zeros of the polynomial $p(z) = z^9 + 2z^5 - 2z^4 + z + 3$ in the right half-plane.
5. For a fixed real number α , find the number of zeros of $z^4 + z^3 + 4z^2 + \alpha z + 3$ satisfying $\operatorname{Re} z < 0$. (Your answer depends on α .)
6. For a fixed real number α , find the number of solutions of $z^5 + 2z^3 - z^2 + z = \alpha$ satisfying $\operatorname{Re} z > 0$.
7. For a fixed complex number λ , show that if m and n are large integers, then the equation $e^z = z + \lambda$ has exactly $m + n$ solutions in the horizontal strip $\{-2\pi im < \operatorname{Im} z < 2\pi in\}$.
8. Show that if $\operatorname{Re} \lambda > 1$, then the equation $e^z = z + \lambda$ has exactly one solution in the left half-plane.
9. Show that if $f(z)$ is analytic in a domain D , and if γ is a closed curve in D such that the values of $f(z)$ on γ lie in the slit plane

$\mathbb{C} \setminus (-\infty, 0]$, then the increase in the argument of $f(z)$ around γ is zero.

2. Rouché's Theorem

There is a general principle to the effect that the number of zeros of an analytic function on a domain does not change if we make a small change in the function. In other words, the number of zeros of an analytic function remains constant under small perturbations. One version of this principle is as follows.

Theorem (Rouché's Theorem). *Let D be a bounded domain with piecewise smooth boundary ∂D . Let $f(z)$ and $h(z)$ be analytic on $D \cup \partial D$. If $|h(z)| < |f(z)|$ for $z \in \partial D$, then $f(z)$ and $f(z) + h(z)$ have the same number of zeros in D , counting multiplicities.*

The hypothesis implies that $f(z) \neq 0$ on ∂D , and that $f(z) + h(z) \neq 0$ on ∂D . From $f(z) + h(z) = f(z)[1 + h(z)/f(z)]$, we obtain

$$(2.1) \quad \arg(f(z) + h(z)) = \arg(f(z)) + \arg\left(1 + \frac{h(z)}{f(z)}\right).$$

Since $|h(z)/f(z)| < 1$, the values of $1 + h(z)/f(z)$ lie in the right half-plane, and the increase of $\arg(1 + h(z)/f(z))$ around a closed boundary curve is 0. From (2.1) we see that $\arg(f(z) + h(z))$ and $\arg f(z)$ have the same increase around ∂D . By the argument principle, the functions have the same number of zeros in D .

Example. We use Rouché's theorem to find the number of zeros of the polynomial $p(z) = z^6 + 9z^4 + z^3 + 2z + 4$ that lie inside the unit circle. To apply Rouché's theorem, we seek to express $p(z)$ in the form

$$p(z) = \text{BIG} + \text{little},$$

where the function "BIG" dominates "little" on the unit circle, and where it is apparent how many zeros "BIG" has inside the unit circle. Our choice for "BIG" in this case is $f(z) = 9z^4$, which has four zeros inside the unit circle, all at the origin. The function "little" is $h(z) = z^6 + z^3 + 2z + 4$, which satisfies $|h(z)| < |f(z)|$ for $|z| = 1$. By Rouché's theorem, $p(z) = f(z) + h(z)$ also has four zeros inside the unit circle.

Example. We wish to find all solutions of the equation $e^z = 1 + 2z$ that satisfy $|z| < 1$. One obvious solution is $z = 0$. To determine whether there are other solutions, we apply Rouché's theorem to count the number of zeros of $e^z - 1 - 2z$ in the unit disk. In this case, the term $f(z) = -2z$ is the dominant term on the unit circle. We take $h(z) = e^z - 1$, and we

estimate

$$|h(z)| = |e^z - 1| = \left| z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots \right| \leq |z| + \frac{|z|^2}{2!} + \frac{|z|^3}{3!} + \cdots.$$

If $|z| = 1$, the right-hand side is $e - 1 \approx 1.7 < 2$. Thus $|h(z)| < |f(z)|$ for $|z| = 1$. Since $f(z)$ has only one zero in the unit disk, $f(z) + h(z)$ also has only one zero, and the equation has no solutions in the unit disk other than $z = 0$.

There is a simple proof of the fundamental theorem of algebra based on Rouché's theorem. Let

$$p(z) = z^m + a_{m-1}z^{m-1} + \cdots + a_0$$

be a monic polynomial of degree $m \geq 1$. For $|z|$ large, the term z^m dominates. We take $f(z) = z^m$ and $h(z) = a_{m-1}z^{m-1} + \cdots + a_0$. If R is large, we have $|h(z)| < |f(z)|$ for $|z| = R$. Consequently, $p(z) = f(z) + h(z)$ has the same number of zeros in the disk $|z| < R$ as $f(z) = z^m$, which is m .

Exercises for VIII.2

1. Show that $2z^5 + 6z - 1$ has one root in the interval $0 < x < 1$ and four roots in the annulus $\{1 < |z| < 2\}$.
2. How many roots does $z^9 + z^5 - 8z^3 + 2z + 1$ have between the circles $\{|z| = 1\}$ and $\{|z| = 2\}$?
3. Show that if m and n are positive integers, then the polynomial

$$p(z) = 1 + z + \frac{z^2}{2!} + \cdots + \frac{z^m}{m!} + 3z^n$$

has exactly n zeros in the unit disk.

4. Fix a complex number λ such that $|\lambda| < 1$. For $n \geq 1$, show that $(z - 1)^n e^z - \lambda$ has n zeros satisfying $|z - 1| < 1$ and no other zeros in the right half-plane. Determine the multiplicity of the zeros.
5. For a fixed λ satisfying $|\lambda| < 1$, show that $(z - 1)^n e^z + \lambda(z + 1)^n$ has n zeros in the right half-plane, which are all simple if $\lambda \neq 0$.
6. Let $p(z) = z^6 + 9z^4 + z^3 + 2z + 4$ be the polynomial treated in the example in this section. (a) Determine which quadrants contain the four zeros of $p(z)$ that lie inside the unit circle. (b) Determine which quadrants contain the two zeros of $p(z)$ that lie outside the unit circle. (c) Show that the two zeros of $p(z)$ that lie outside the unit circle satisfy $\{|z \pm 3i| < 1/10\}$.
7. Let $f(z)$ and $g(z)$ be analytic functions on the bounded domain D that extend continuously to ∂D and satisfy $|f(z) + g(z)| < |f(z)| + |g(z)|$ on ∂D . Show that $f(z)$ and $g(z)$ have the same number of

zeros in D , counting multiplicity. *Remark.* This is a variant of Rouché's theorem, in which the hypotheses are symmetric in $f(z)$ and $g(z)$. Rouché's theorem is obtained by setting $h(z) = -f(z) - g(z)$. For the solution of the exercise, see Exercise 9 in the preceding section.

8. Let D be a bounded domain, and let $f(z)$ and $h(z)$ be meromorphic functions on D that extend to be analytic on ∂D . Suppose that $|h(z)| < |f(z)|$ on ∂D . Show by example that $f(z)$ and $f(z) + h(z)$ can have different numbers of zeros on D . What *can* be said about $f(z)$ and $f(z) + h(z)$? Prove your assertion.
9. Let $f(z)$ be a continuously differentiable function on a domain D . Suppose that for all complex constants a and b , the increase in the argument of $f(z) + az + b$ around any small circle in D on which $f(z) + az + b \neq 0$ is nonnegative. Show that $f(z)$ is analytic.

3. Hurwitz's Theorem

The argument principle (logarithmic integral) provides a tool for studying the behavior of the zeros of a convergent sequence of analytic functions. The zeros of the functions in the sequence converge to the zeros of the limit function, in a sense made precise by the following theorem.

Theorem (Hurwitz's Theorem). *Suppose $\{f_k(z)\}$ is a sequence of analytic functions on a domain D that converges normally on D to $f(z)$, and suppose that $f(z)$ has a zero of order N at z_0 . Then there exists $\rho > 0$ such that for k large, $f_k(z)$ has exactly N zeros in the disk $\{|z - z_0| < \rho\}$, counting multiplicity, and these zeros converge to z_0 as $k \rightarrow \infty$.*

For the proof, let $\rho > 0$ be sufficiently small so that the closed disk $\{|z - z_0| \leq \rho\}$ is contained in D and so that $f(z) \neq 0$ for $0 < |z - z_0| \leq \rho$. Choose $\delta > 0$ such that $|f(z)| \geq \delta$ on the circle $|z - z_0| = \rho$. Since $f_k(z)$ converges uniformly to $f(z)$ for $|z - z_0| \leq \rho$, for k large we have $|f_k(z)| > \delta/2$ for $|z - z_0| = \rho$, and further, $f'_k(z)/f_k(z)$ converges uniformly to $f'(z)/f(z)$ for $|z - z_0| = \rho$. Hence the integrals converge,

$$\frac{1}{2\pi i} \int_{|z-z_0|=\rho} \frac{f'_k(z)}{f_k(z)} dz \longrightarrow \frac{1}{2\pi i} \int_{|z-z_0|=\rho} \frac{f'(z)}{f(z)} dz.$$

Now, the integral on the left is the number N_k of zeros of $f_k(z)$ in the disk $\{|z - z_0| < \rho\}$, while the integral on the right is the number N of zeros of $f(z)$ in the disk. Since $N_k \rightarrow N$, in fact $N_k = N$ for k large, that is, $f_k(z)$ has N zeros satisfying $|z - z_0| < \rho$. The same argument works for smaller $\rho > 0$, so the zeros of $f_k(z)$ must accumulate at z_0 .

An alternative proof of Hurwitz's theorem can be based on Rouché's theorem, applied to the "little" function $f_k(z) - f(z)$ and the "big" function $f(z)$. See Exercise 2.

We say that a function is **univalent** on a domain D if it is analytic and one-to-one on D . In other words, the univalent functions on D are the conformal maps of D to other domains. If we apply the preceding theorem to univalent functions, where $N = 1$, we obtain the following, which is sometimes also referred to as Hurwitz's theorem.

Theorem. *Suppose $\{f_k(z)\}$ is a sequence of univalent functions on a domain D that converges normally on D to a function $f(z)$. Then either $f(z)$ is univalent or $f(z)$ is constant.*

Indeed, suppose the limit function $f(z)$ is not constant. Suppose z_0 and ζ_0 satisfy $f(z_0) = f(\zeta_0) = w_0$. Then z_0 and ζ_0 are zeros of finite order for $f(z) - w_0$. By the preceding theorem, there are sequences $z_k \rightarrow z_0$ and $\zeta_k \rightarrow \zeta_0$ such that $f_k(z_k) - w_0 = 0$, and $f_k(\zeta_k) - w_0 = 0$. Since $f_k(z)$ is univalent, we have $z_k = \zeta_k$, and in the limit $z_0 = \zeta_0$. Thus $f(z)$ is univalent.

Example. For each $k \geq 1$, the function $f_k(z) = z/k$ is univalent on the complex plane. The sequence $\{f_k(z)\}$ converges normally to the constant function $f(z) \equiv 0$. Thus a normal limit of univalent functions need not be univalent, and both alternatives of the theorem occur.

Exercises for VIII.3

1. Let $\{f_k(z)\}$ be a sequence of analytic functions on D that converges normally to $f(z)$, and suppose that $f(z)$ has a zero of order N at $z_0 \in D$. Use Rouché's theorem to show that there exists $\rho > 0$ such that for k large, $f_k(z)$ has exactly N zeros counting multiplicity on the disk $\{|z - z_0| < \rho\}$.
2. Let $\mathcal{S}$ be the family of univalent functions $f(z)$ defined on the open unit disk $\{|z| < 1\}$ that satisfy $f(0) = 0$ and $f'(0) = 1$. Show that $\mathcal{S}$ is closed under normal convergence, that is, if a sequence in $\mathcal{S}$ converges normally to $f(z)$, then $f \in \mathcal{S}$. *Remark.* It is also true, but more difficult to prove, that $\mathcal{S}$ is a compact family of analytic functions, that is, every sequence in $\mathcal{S}$ has a normally convergent subsequence.

4. Open Mapping and Inverse Function Theorems

Let $f(z)$ be a meromorphic function on a domain D . We say that $f(z)$ **attains the value w_0 m times at z_0** if $f(z) - w_0$ has a zero of order m at z_0 . We make the usual modifications to cover the cases $z_0 = \infty$ and

$w_0 = \infty$, so that $f(z)$ attains a finite value w_0 m times at $z_0 = \infty$ if $f(1/z) - w_0$ has a zero of order m at $z = 0$, and $f(z)$ attains the value ∞ m times at z_0 if z_0 is a pole of $f(z)$ of order m .

The number of times that $f(z)$ attains a value w_0 on D is obtained by adding the number of times it attains the value w_0 at the various points of D .

Example. The polynomial $z^m + 1$ attains the value $w = 1$ m times at $z = 0$, and it attains the value $w = \infty$ m times at $z = \infty$.

Example. A polynomial $f(z)$ of degree $m \geq 1$ attains each value $w \in \mathbb{C}^*$ m times on $\mathbb{C}^*$, always counting multiplicity. In fact, for fixed finite w , $f(z) - w$ is a polynomial of degree m , which has m zeros counting multiplicity. Since $f(z)$ has a pole of order m at ∞ , it attains the value $w = \infty$ m times at $z = \infty$.

We can use the logarithmic integral to study the dependence on w of the number of times that an analytic or meromorphic function attains the value w . The technique is the same as that used in the preceding section, except that instead of a sequence of functions we treat functions that depend continuously on a parameter. The basic idea is as follows.

Let $f(z)$ be a nonconstant analytic function on a domain D . Let $z_0 \in D$, $w_0 = f(z_0)$, and assume that $f(z) - w_0$ has a zero of order m at z_0 . Since the zeros of $f(z) - w_0$ are isolated, we can choose $\rho > 0$ so that $f(z) - w_0 \neq 0$ for $0 < |z - z_0| \leq \rho$. Let $\delta > 0$ satisfy $|f(z) - w_0| \geq \delta$ for $|z - z_0| = \rho$. The integral

$$(4.1) \quad N(w) = \frac{1}{2\pi i} \int_{|z-z_0|=\rho} \frac{f'(z)}{f(z)-w} dz, \quad |w - w_0| < \delta,$$

is then defined, and it depends analytically on w . Since $N(w)$ is the number of zeros of $f(z) - w$ in the disk $\{|z - z_0| < \rho\}$, the analytic function $N(w)$ is integer-valued, hence constant. Since $N(w_0) = m$, we obtain $N(w) = m$ for $|w - w_0| < \delta$. Thus each value w satisfying $|w - w_0| < \delta$ is assumed exactly m times, counting multiplicity, by $f(z)$ in the disk $\{|z - z_0| < \rho\}$.

As a first consequence of this analysis, we show that nonconstant analytic functions are “open mappings.”

Theorem (Open Mapping Theorem for Analytic Functions). *If $f(z)$ is analytic on a domain D , and $f(z)$ is not constant, then $f(z)$ maps open sets to open sets, that is, $f(U)$ is open for each open subset U of D .*

Indeed, let $w_0 \in f(U)$, say $w_0 = f(z_0)$. If we apply the above discussion to $f(z)$, we obtain a disk centered at z_0 of radius ρ and contained in U , such that the values of $f(z)$ on this disk include the disk centered at w_0 of radius δ . This implies that $f(U)$ is open.

Now consider the case where $f(z) - w_0$ has a simple zero at z_0 . In this case, $N(w_0) = N_0 = 1$, and we conclude that $N(w) = 1$ for $|w - w_0| < \delta$. Thus every value w satisfying $|w - w_0| < \delta$ is taken on exactly once by $f(z)$ in the disk $\{|z - z_0| < \rho\}$. This allows us to define an inverse function $f^{-1}(w)$ for $|w - w_0| < \delta$. Further, the residue theorem allows us to derive an explicit formula for the inverse function.

Theorem (Inverse Function Theorem). *Suppose $f(z)$ is analytic for $|z - z_0| \leq \rho$ and satisfies $f(z_0) = w_0$, $f'(z_0) \neq 0$, and $f(z) \neq w_0$ for $0 < |z - z_0| \leq \rho$. Let $\delta > 0$ be chosen such that $|f(z) - w_0| \geq \delta$ for $|z - z_0| = \rho$. Then for each w such that $|w - w_0| < \delta$, there is a unique z satisfying $|z - z_0| < \rho$ and $f(z) = w$. Writing $z = f^{-1}(w)$, we have*

$$(4.2) \quad f^{-1}(w) = \frac{1}{2\pi i} \int_{|\zeta - z_0| = \rho} \frac{\zeta f'(\zeta)}{f(\zeta) - w} d\zeta, \quad |w - w_0| < \delta.$$

It remains only to establish (4.2). Fix w such that $|w - w_0| < \delta$, and set $z = f^{-1}(w)$, so that $f(z) = w$. The function $\zeta f'(\zeta)/[f(\zeta) - w]$ is an analytic function of ζ for $|\zeta - z_0| \leq \rho$, except for a simple pole at $\zeta = z$ with residue

$$\operatorname{Res} \left[\frac{\zeta f'(\zeta)}{f(\zeta) - w}, z \right] = \lim_{\zeta \rightarrow z} \frac{(\zeta - z)\zeta f'(\zeta)}{f(\zeta) - w} = z.$$

The residue theorem yields (4.2) immediately.

We remark that this gives an alternative proof of the existence of a locally defined inverse for $f(z)$ when $f'(z) \neq 0$, which depends on the residue theorem rather than on the inverse function theorem. This procedure also gives an explicit disk on which the inverse function is defined. The explicit formula (4.2) shows that the inverse function is analytic.

Exercises for VIII.4

1. Suppose D is a bounded domain with piecewise smooth boundary. Let $f(z)$ be meromorphic and $g(z)$ analytic on D . Suppose that both $f(z)$ and $g(z)$ extend analytically across the boundary of D , and that $f(z) \neq 0$ on ∂D . Show that

$$\frac{1}{2\pi i} \oint_{\partial D} g(z) \frac{f'(z)}{f(z)} dz = \sum_{j=1}^n m_j g(z_j),$$

where $z_1, \dots, z_n$ are the zeros and poles of $f(z)$, and m_j is the order of $f(z)$ at z_j .

2. Let $f(z)$ be a meromorphic function on the complex plane that is doubly periodic. Suppose that the zeros and poles of $f(z)$ are at the points $z_1, \dots, z_n$ and at their translates by periods of $f(z)$, and

suppose no z_j is a translate by a period of another z_k . Let m_j be the order of $f(z)$ at z_j . Show that $\sum m_j z_j$ is a period of $f(z)$. *Hint.* Integrate $zf'(z)/f(z)$ around the boundary of the fundamental parallelogram P constructed in Section VI.5.

3. Let $\{f_k(z)\}$ be a sequence of analytic functions on a domain D that converges normally to $f(z)$. Suppose that $f_k(z)$ attains each value w at most m times (counting multiplicity) in D . Show that either $f(z)$ is constant, or $f(z)$ attains each value w at most m times in D .
4. Let $f(z)$ be an analytic function on the open unit disk $\mathbb{D} = \{|z| < 1\}$. Suppose there is an annulus $U = \{r < |z| < 1\}$ such that the restriction of $f(z)$ to U is one-to-one. Show that $f(z)$ is one-to-one on $\mathbb{D}$.
5. Let $f(z) = p(z)/q(z)$ be a rational function, where $p(z)$ and $q(z)$ are polynomials that are relatively prime (no common zeros). We define the **degree** of $f(z)$ to be the larger of the degrees of $p(z)$ and $q(z)$. Denote the degree of $f(z)$ by d . (a) Show that each value $w \in \mathbb{C}$, $w \neq f(\infty)$, is assumed d times by $f(z)$ on $\mathbb{C}$. (b) Show that $f(z)$ attains each value $w \in \mathbb{C}^*$ d times on $\mathbb{C}^*$ (as always, counting multiplicity).
6. Let $f(z)$ be a meromorphic function on the complex plane, and suppose there is an integer m such that $f^{-1}(w)$ has at most m points for all $w \in \mathbb{C}$. Show that $f(z)$ is a rational function.
7. Let $F(z, w)$ be a continuous function of z and w that depends analytically on z for each fixed w , and let $F_1(z, w)$ denote the derivative of $F(z, w)$ with respect to z . Suppose $F(z_0, w_0) = 0$, and $F_1(z_0, w_0) \neq 0$. Choose ρ such that $F(z, w_0) \neq 0$ for $0 < |z - z_0| \leq \rho$. (a) Show that there exists $\delta > 0$ such that if $|w - w_0| < \delta$, there is a unique $z = g(w)$ satisfying $|z - z_0| < \rho$ and $F(z, w) = 0$. (b) Show that

$$g(w) = \frac{1}{2\pi i} \int_{|\zeta - z_0| = \rho} \frac{\zeta F_1(\zeta, w)}{F(\zeta, w)} d\zeta, \quad |w - w_0| < \delta.$$

- (c) Suppose further that $F(z, w)$ is analytic in w for each fixed z , and let $F_2(z, w)$ denote the derivative of $F(z, w)$ with respect to w . Show that $g(w)$ is analytic, and

$$g'(w) = -F_2(g(w), w)/F_1(g(w), w).$$

- (d) Derive the inverse function theorem given in this section, together with the usual formula for the derivative of the inverse function, as a corollary of (a), (b), and (c).

Remark. This is the **implicit function theorem for analytic functions**. Note that a specific formula is given for the function $g(w)$ defined implicitly by $F(g(w), w) = 0$.

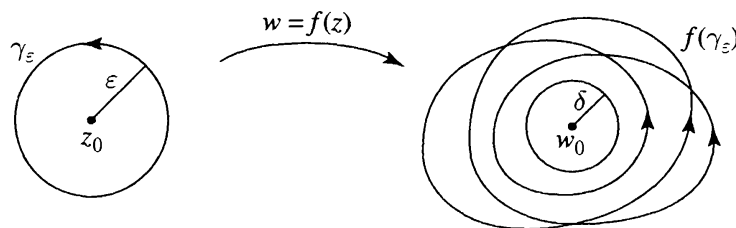
8. Let D be a bounded domain, and let $f(z)$ be a continuous function on $D \cup \partial D$ that is analytic on D . Show that $\partial(f(D)) \subseteq f(\partial D)$, that is, the boundary of the open set $f(D)$ is contained in the image under $f(z)$ of the boundary of D .

5. Critical Points

Let $f(z)$ be a nonconstant analytic function on a domain D . A point z_0 is called a **critical point** of $f(z)$ if $f'(z_0) = 0$. The value $f(z_0) = w_0$ is called a **critical value** of $f(z)$. We define the **order** of the critical point z_0 to be the order of zero of $f'(z)$ at z_0 . Since the critical points are the zeros of the nonconstant analytic function $f'(z)$, critical points are isolated. Our aim is to understand the behavior of an analytic function near a critical point, and to understand the behavior of the inverse function near the corresponding critical value.

Example. The function $f(z) = z^m + c$ has a critical point of order $m - 1$ at $z = 0$, with critical value c . The inverse function for $f(z)$ is given by the m branches of $(w - c)^{1/m}$.

Suppose z_0 is a critical point of $f(z)$ of order $m - 1$ with critical value w_0 . Then $f(z) - w_0$ has a zero of order m at z_0 , and we can proceed as before by considering the logarithmic integral (4.1). This time the constant value of $N(w)$ is m . Thus for $\rho > 0$ small and for w near w_0 , there are m points $z_1(w), \dots, z_m(w)$, repeated according to multiplicity, such that $f(z)$ attains the value w in the disk $\{|z - z_0| < \rho\}$ precisely at the points $z_1(w), \dots, z_m(w)$. We wish to see how the points $z_j(w)$ depend on w . To do this, we will use a simple procedure, based only on the existence of an analytic inverse for an analytic function at a noncritical point. The procedure is to make a change of variable in order to show that the behavior of $f(z)$ near z_0 is effectively the same as the behavior of the analytic function ζ^m at the critical point $\zeta = 0$. We proceed as follows.

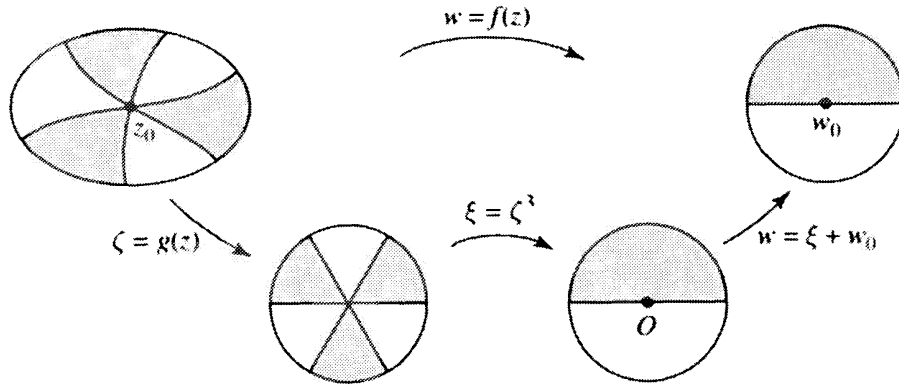


Since $f(z) - w_0$ has order m at z_0 , we can factor $f(z) - w_0 = (z - z_0)^m h(z)$, where $h(z)$ is analytic at z_0 , and $h(z_0) \neq 0$. Near z_0 we define an analytic

branch of $h(z)^{1/m}$, and then for $g(z) = (z - z_0)h(z)^{1/m}$ we have

$$f(z) - w_0 = g(z)^m, \quad f(z) = g(z)^m + w_0.$$

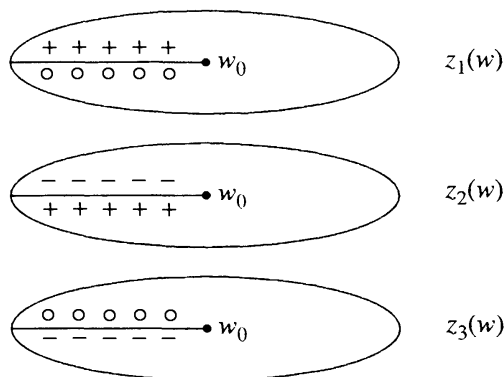
Moreover, $g(z)$ is analytic at z_0 and has a simple zero there, so $g'(z_0) \neq 0$ and $g(z)$ is one-to-one near z_0 . Thus $f(z)$ is represented as the composition of three functions, the univalent function $g(z)$, followed by the map $\zeta \mapsto \zeta^m$, followed by a translation $\xi \mapsto \xi + w_0$. Since we understand the power function $\xi = \zeta^m$ and the translation $w = \xi + w_0$ very well, and since $g(z)$ is univalent near z_0 , we can draw from this a clear picture of the behavior of $f(z)$. For instance, the set where ζ^m is real consists of m straight line segments passing through 0, dividing the ζ -plane near $\zeta = 0$ into $2m$ sectors of aperture π/m , on which $\text{Im}(\zeta^m)$ is alternately positive and negative. The set where $f(z) - w_0$ is real is the image under $g^{-1}(\zeta)$ of these line segments and hence consists of m “analytic” curves through z_0 . They divide the z -plane near z_0 into $2m$ pie-slice domains with vertex at z_0 and aperture π/m , on which $\text{Im}(f(z) - w_0)$ is alternately positive and negative.



A point w near w_0 , $w \neq w_0$, has m distinct preimages $z_1(w), \dots, z_m(w)$. They are the m branches of $(w - w_0)^{1/m}$ composed with $g^{-1}(\zeta)$. If we make a branch cut and consider the principal branch $(w - w_0)^{1/m}$ on the slit disk $\{|w - w_0| < \delta\} \setminus (w_0 - \delta, w_0]$, the other branches are of the form $e^{2\pi ij/m}(w - w_0)^{1/m}$, and the preimages of w are given by the composition of $g^{-1}(\zeta)$ and these branches,

$$z_j(w) = g^{-1} \left(e^{2\pi ij/m} (w - w_0)^{1/m} \right), \quad 1 \leq j \leq m.$$

In particular, for $w \neq w_0$, the preimages $z_j(w)$ are distinct, and each $z_j(w)$ depends analytically on w . The values of $z_j(w)$ at the “top edge” of the slit coincide with the values of $z_{j+1}(w)$ at the “bottom edge” of the slit, so that $z_j(w)$ is continued analytically to $z_{j+1}(w)$ when w follows a circle around w_0 in the positive direction. In the same way that we constructed a Riemann surface for the inverse function of ζ^m , we can construct a Riemann surface for the inverse function $z(w)$ of $w = f(z)$. We create m copies of the slit disk, and we define $z(w)$ to be $z_j(w)$ on the j th copy. Then we glue



together the edges of the branch cuts so that the function $z(w)$ becomes continuous on the surface.

Now consider the function

$$(z - z_1(w)) \cdots (z - z_m(w)) = z^m + a_{m-1}(w)z^{m-1} + \cdots + a_1(w)z + a_0(w),$$

where

$$\begin{aligned} a_0(w) &= (-1)^m z_1(w) \cdots z_m(w), \\ &\vdots \\ a_{m-1}(w) &= - \sum_{j=1}^m z_j(w). \end{aligned}$$

If we continue $a_k(w)$ around a circle centered at w_0 , the values of the $z_j(w)$'s are permuted and we return to the same value for $a_k(w)$. Thus $a_k(w)$ is analytic in the punctured disk $0 < |w - w_0| < \varepsilon$. Evidently, $a_k(w)$ is bounded as $w \rightarrow w_0$. By Riemann's theorem on removable singularities, $a_k(w)$ is analytic at w_0 . In fact, the value $a_k(w_0)$ is just the coefficient of z^k in the expansion of $(z - z_0)^m$ in powers of z . Thus we have found a polynomial equation

$$(5.1) \quad z^m + a_{m-1}(w)z^{m-1} + \cdots + a_1(w)z + a_0(w) = 0,$$

with analytic coefficients $a_0(w), \dots, a_{m-1}(w)$, that reduces to the equation $(z - z_0)^m = 0$ at $w = w_0$ and that has solutions given precisely by $z_1(w), \dots, z_m(w)$.

We summarize our results as follows.

Theorem. *Let $f(z)$ be analytic at z_0 . Suppose z_0 is a critical point of order $m - 1$ for $f(z)$, with critical value $f(z_0) = w_0$. Let $\rho > 0$ satisfy $f(z) \neq w_0$ for $0 < |z - z_0| \leq \rho$, and let $\delta > 0$ satisfy $|f(z) - w_0| \geq \delta$ for $|z - z_0| = \rho$. Then for each w such that $0 < |w - w_0| < \delta$, the equation $f(z) = w$ has exactly m distinct solutions $z_1(w), \dots, z_m(w)$ in the disk $\{|z - z_0| < \rho\}$. The functions $z_1(w), \dots, z_m(w)$ can be chosen to depend analytically on w in the slit disk $\{|w - w_0| < \delta\} \setminus (w_0 - \delta, w_0]$. They form the m branches of an analytic function $z = z(w)$ on an m -sheeted Riemann*

surface over the punctured disk $\{0 < |w - w_0| < \delta\}$. The $z_j(w)$'s are the solutions of a polynomial equation (5.1) with coefficients $a_j(w)$ that are analytic for $|w - w_0| < \delta$.

We have dealt with the solutions of the equation $f(z) - w = 0$ near a solution point (z_0, w_0) . A similar state of affairs holds for the solutions of an equation $F(z, w) = 0$, where $F(z, w)$ depends analytically on both z and w in a neighborhood of a solution point (z_0, w_0) , though in this case the solution set is a finite union of surfaces passing through (z_0, w_0) . This is the simplest example of an **analytic variety**, which is defined to be the set of common zeros of a family of analytic functions on some domain in complex n -space.

Exercises for VIII.5

1. Find the critical points and critical values of $f(z) = z + 1/z$. Sketch the curves where $f(z)$ is real. Sketch the regions where $\operatorname{Im} f(z) > 0$ and where $\operatorname{Im} f(z) < 0$.
2. Suppose $g(z)$ is analytic at $z = 0$, with power series $g(z) = 2 + iz^4 + \mathcal{O}(z^5)$. Sketch and label the curves passing through $z = 0$ where $\operatorname{Re} g(z) = 2$ and where $\operatorname{Im} g(z) = 0$.
3. Find the critical points and critical values of $f(z) = z^2 + 1$. Sketch the set of points z such that $|f(z)| \leq 1$, and locate the critical points of $f(z)$ on the sketch.
4. Suppose that $f(z)$ is analytic at z_0 . Show that if the set of z such that $\operatorname{Re} f(z) = \operatorname{Re} f(z_0)$ consists of just one curve passing through z_0 , then $f'(z_0) \neq 0$. Show also that if the set of z such that $|f(z)| = |f(z_0)|$ consists of just one curve passing through z_0 , then $f'(z_0) \neq 0$.
5. How many critical points, counting multiplicity, does a polynomial of degree m have in the complex plane? Justify your answer.
6. Find and plot the critical points and critical values of $f(z) = z^2 + 1$ and of its iterates $f(f(z)) = (z^2 + 1)^2 + 1$ and $f(f(f(z)))$. *Suggestion.* Use the chain rule.
7. Let $f(z)$ be a polynomial of degree m . How many (finite) critical points does the N -fold iterate $f \circ \cdots \circ f$ (N times) have? Describe them in terms of the critical points of $f(z)$.
8. We define a pole z_0 of $f(z)$ to be a **critical point of $f(z)$ of order k** if z_0 is a critical point of $1/f(z)$ of order k . We define $z = \infty$ to be a **critical point of $f(z)$ of order k** if $w = 0$ is a critical point of $g(w) = f(1/w)$ of order k . Show that with this definition, a point

$z_0 \in \mathbb{C}^*$ is a critical point of order k for a meromorphic function $f(z)$ if and only if there are open sets U containing z_0 and V containing $w_0 = f(z_0)$ such that each $w \in V$, $w \neq w_0$, has exactly $k + 1$ preimages in U . *Remark.* We say that $f(z)$ is a $(k + 1)$ -**sheeted covering of $f^{-1}(V \setminus \{w_0\}) \cap U$ over $V \setminus \{w_0\}$** .

9. Show that a polynomial of degree m , regarded as a meromorphic function on $\mathbb{C}^*$, has a critical point of order $m - 1$ at $z_0 = \infty$.
10. Locate the critical points and critical values in the extended complex plane of the polynomial $f(z) = z^4 - 2z^2$. Determine the order of each critical point. Sketch the set of points z such that $\operatorname{Im} f(z) \geq 0$.
11. Show that if f is a rational function, and if g is a fractional linear transformation, then f and $g \circ f$ have the same critical points in the extended complex plane $\mathbb{C}^*$. What can be said about the critical values of $g \circ f$? What can be said about the critical points and critical values of $f \circ g$?
12. Let $f(z) = p(z)/q(z)$ be a rational function of degree d , so that $p(z)$ and $q(z)$ are relatively prime, and d is the larger of the degrees of $p(z)$ and $q(z)$. (See Exercise 4.5.) Show that $f(z)$ has $2d - 2$ critical points, counting multiplicity, in the extended complex plane $\mathbb{C}^*$. *Hint.* If $\deg p \neq \deg q$, then the number of critical points of $f(z)$ in the finite plane $\mathbb{C}$ is $\deg(qp' - q'p) = \deg p + \deg q - 1$, while the order of the critical point at ∞ is $|\deg p - \deg q| - 1$.
13. Show that the set of solution points (w, z) of the equation $z^2 - 2(\cos w)z + 1 = 0$ consists of the graphs of two entire functions $z_1(w)$ and $z_2(w)$ of w . Specify the entire functions, and determine where their graphs meet. *Remark.* The solution set forms a reducible one-dimensional analytic variety in $\mathbb{C}^2$.
14. Let $a_0(w), \dots, a_{m-1}(w)$ be analytic in a neighborhood of $w = 0$ and vanish at $w = 0$. Consider the monic polynomial in z whose coefficients are analytic functions in w ,

$$P(z, w) = z^m + a_{m-1}(w)z^{m-1} + \cdots + a_0(w), \quad |w| < \delta.$$

Suppose that for each fixed w , $0 < |w| < \delta$, there are m distinct solutions of $P(z, w) = 0$.

- (a) Show that the m roots of the equation $P(z, w) = 0$ determine analytic functions $z_1(w), \dots, z_m(w)$ in the slit disk $\{|w| < \delta\} \setminus (-\delta, 0]$. *Hint.* Use the implicit function theorem (Exercise 4.7).
- (b) Glue together branch cuts to form an m -sheeted Riemann surface over the punctured disk on which the branches $z_j(w)$ determine a continuous (single-valued, analytic) function.

- (c) Suppose that the indices are arranged so that for some fixed k , $1 \leq k \leq n$, the continuation of $z_j(w)$ once around $w = 0$ is $z_{j+1}(w)$ for $1 \leq j \leq k-1$, while the continuation of $z_k(w)$ once around $w = 0$ is $z_1(w)$. Show that $Q(z, w) = (z - z_1(w)) \cdots (z - z_k(w))$ determines a polynomial in z whose coefficients are analytic functions of w for $|w| < \delta$. Show further that the polynomial $Q(z, w)$ is an irreducible factor of $P(z, w)$, and that all irreducible factors of $P(z, w)$ arise from subsets of the $z_j(w)$'s in this way.
- (d) Show that if $f(z)$ is an analytic function that has a zero of order m at $z = 0$, and $z_1(w), \dots, z_m(w)$ are the solutions of the equation $w = f(z)$, then the polynomial $P(z, w) = (z - z_1(w)) \cdots (z - z_m(w))$ is irreducible.

15. Consider monic polynomials in z of the form

$$P(z, w) = z^m + a_{m-1}(w)z^{m-1} + \cdots + a_0(w),$$

where the functions $a_0(w), \dots, a_{m-1}(w)$ are defined and meromorphic in some disk centered at $w = 0$. Let $P_0(z, w)$ and $P_1(z, w)$ be two such polynomials, and consider the following algorithm. Using the division algorithm, find polynomials $A_2(z, w)$ and $P_2(z, w)$ such that $P_0(z, w) = A_2(z, w)P_1(z, w) + P_2(z, w)$ and the degree of $P_2(z, w)$ is less than the degree of $P_1(z, w)$. Continue in this fashion, finding polynomials $A_{j+1}(z, w)$ and $P_{j+1}(z, w)$ such that

$$P_{j-1}(z, w) = A_{j+1}(z, w)P_j(z, w) + P_{j+1}(z, w)$$

and $\deg P_{j+1}(z, w) < \deg P_j(z, w)$, until eventually we reach

$$P_{\ell+1}(z, w) = 0, \quad P_{\ell-1}(z, w) = A_{\ell}(z, w)P_{\ell}(z, w).$$

Let $D(z, w)$ be the monic polynomial in z obtained by dividing $P_{\ell}(z, w)$ by the coefficient of the highest power of z .

- (a) Show that $D(z, w)$ is the greatest common divisor of $P_0(z, w)$ and $P_1(z, w)$, in the sense that $D(z, w)$ divides both $P_0(z, w)$ and $P_1(z, w)$, and each polynomial that divides both $P_0(z, w)$ and $P_1(z, w)$ also divides $D(z, w)$.
- (b) Show that there are polynomials $A(z, w)$ and $B(z, w)$ such that $D = AP_1 + BP_2$.
- (c) Show that if $P_0(z, w)$ and $P_1(z, w)$ are relatively prime (that is, $D(z, w) = 1$), then there is $\varepsilon > 0$ such that for each fixed w , $0 < |w| < \varepsilon$, the polynomials $P_0(z, w)$ and $P_1(z, w)$ have no common zeros.
- (d) Show that any polynomial $P(z, w)$ as above can be factored as a product of irreducible polynomials, and the factorization is unique up to the order of the factors and multiplication of a factor by a meromorphic function in w .

- (e) Show that if the coefficients of $P(z, w)$ are analytic at $w = 0$, then the coefficients of each of the irreducible factors of $P(z, w)$ are analytic at $w = 0$.
- (f) Show that if $P(z, w)$ is irreducible, then there is $\varepsilon > 0$ such that for each fixed w , $0 < |w| < \varepsilon$, the roots of $P(z, w)$ are distinct.
- (g) Show that the results of Exercise 14(a)–(d) hold without the supposition that the solutions of $P(z, w) = 0$ are distinct.

6. Winding Numbers

Let $\gamma(t)$, $a \leq t \leq b$, be a closed path in D . We define the **trace** of γ to be the image $\Gamma = \gamma([a, b])$ of γ . In this section and the next it will be important to distinguish between the parameterized path γ and its trace Γ .

For $z_0 \notin \Gamma$, we define the **winding number** $W(\gamma, z_0)$ of γ around z_0 to be the increase in the argument of $z - z_0$ around γ , normalized by dividing by 2π . Thus the winding number is the integer

$$(6.1) \quad W(\gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - z_0} = \frac{1}{2\pi} \int_{\gamma} d \arg(z - z_0), \quad z_0 \notin \Gamma.$$

We may think of a cameraman standing at z_0 and videotaping a child emerging through a door, romping around, and returning to the door. The winding number of γ around z_0 is the number of revolutions the cameraman makes as the child runs around the path γ , counting counterclockwise as positive and clockwise as negative.

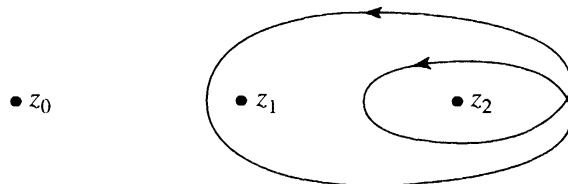
The winding number around z_0 is defined not only for piecewise smooth curves but also for any closed path γ . It is determined by choosing values $h(t)$ of $\arg(\gamma(t) - z_0)$ that vary continuously with t for $a \leq t \leq b$, and setting

$$W(\gamma, z_0) = \frac{1}{2\pi} [h(b) - h(a)].$$

If there is a single-valued determination $\psi(z)$ of $\arg(z - z_0)$ defined on Γ , we can take $h(t) = \psi(\gamma(t))$, and then $h(b) = h(a)$, so that the winding number of γ around z_0 is 0.

Example. The closed curve γ represented by the figure loops twice around z_2 but only once around z_1 , and so $W(\gamma, z_2) = 2$, $W(\gamma, z_1) = 1$. For z_0 lying to the left of the curve, there is a single-valued determination of the argument function defined for $z \in \Gamma$, namely the principal branch $\text{Arg}(z - z_0)$. Consequently, the integrals in (6.1) are zero, and $W(\gamma, z_0) = 0$.

The representation of the winding number as a Cauchy integral in (6.1) shows that $W(\gamma, \zeta)$ depends analytically on ζ for $\zeta \notin \Gamma$. Since the function is integer-valued, it is constant on each component of the complement of Γ . Since the Cauchy integral in (6.1) tends to 0 as $z_0 \rightarrow \infty$, the integer



$W(\gamma, \zeta)$ must be 0 on the unbounded component of the complement of Γ . This can be seen also by noting that there is a single-valued determination of $\arg(z - \zeta)$ defined for $z \in \Gamma$ when ζ is near ∞ . We state our observations as a theorem.

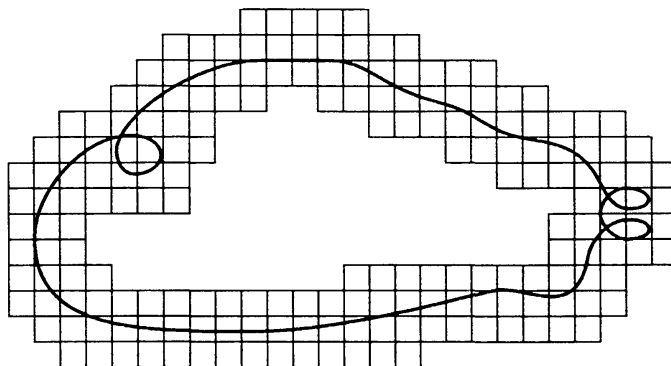
Theorem. *Let $\gamma(t)$, $a \leq t \leq b$, be a closed path in the complex plane, and let $\Gamma = \gamma([a, b])$ be its trace. The winding number $W(\gamma, \zeta)$ is constant on each connected component of $\mathbb{C} \setminus \Gamma$. Further, $W(\gamma, \zeta) = 0$ for all ζ in the unbounded component of $\mathbb{C} \setminus \Gamma$.*

Various of the theorems involving Cauchy integrals around closed curves can be rephrased in terms of winding numbers. We have seen that if D is a star-shaped domain, and if $f(z)$ is analytic on D , then $\int_{\gamma} f(z) dz = 0$ around any closed curve γ in D . This statement extends to arbitrary domains, provided that γ does not wind around a component of the complement of D .

Theorem. *If $f(z)$ is analytic on a domain D , then $\int_{\gamma} f(z) dz = 0$ for each closed path γ in D such that $W(\gamma, \zeta) = 0$ for all $\zeta \in \mathbb{C} \setminus D$.*

The first step of the proof is to find a bounded domain U whose boundary consists of a finite number of piecewise smooth closed curves, such that U and its boundary are contained in D , and such that U contains the trace Γ of γ . To do this, we choose $\delta > 0$ so small that every point of Γ has distance at least 4δ from any point of $\mathbb{C} \setminus D$, and we divide the complex plane into a grid of closed squares of side length δ . (See the figure.) Let K be the union of all the (closed) squares in the grid that contain a point with distance less than δ from Γ , and let $U = K \setminus \partial K$. Then U is a domain with boundary $\partial U = \partial K$ consisting of the sides of squares in K that are adjacent to a square in the grid not in K . By following along these sides, we see that ∂U is a finite union of closed curves consisting of consecutive sides of squares. Two boundary curves may intersect at a point where two squares in U are kitty-corner from each other, but these intersections can be eliminated by adjoining to U a smaller square with vertex in the corner. Finally, we adjoin to U the squares from any component of $\mathbb{C}^* \setminus U$ that does not meet $\mathbb{C}^* \setminus D$. Then each component of $\mathbb{C}^* \setminus U$ meets $\mathbb{C}^* \setminus D$.

Green's theorem is proved (completely rigorously) for domains built in this way from contiguous squares, so that the Cauchy integral theorem is



at our disposal. If $z \in \Gamma$, we have

$$f(z) = \frac{1}{2\pi i} \int_{\partial U} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

If we integrate around γ and then interchange the order of integration, we obtain

$$\int_{\gamma} f(z) dz = \frac{1}{2\pi i} \int_{\partial U} \left[\int_{\gamma} \frac{1}{\zeta - z} dz \right] f(\zeta) d\zeta = -\frac{1}{2\pi i} \int_{\partial U} W(\gamma, \zeta) f(\zeta) d\zeta.$$

The winding number $W(\gamma, \zeta)$ is constant on each connected component of $\mathbb{C} \setminus \Gamma$, and it is zero for $\zeta \in \mathbb{C} \setminus D$. Since each component of $\mathbb{C} \setminus U$ meets $\mathbb{C} \setminus D$, it is zero for $\zeta \in \mathbb{C} \setminus U$, hence also for $\zeta \in \partial U$. Hence the integral is also zero.

As an analogue of the Cauchy integral formula we have the following.

Theorem. *Let $f(z)$ be analytic on a domain D , and let γ be a closed path in D with trace $\Gamma = \gamma([a, b])$. If $W(\gamma, \zeta) = 0$ for all $\zeta \in \mathbb{C} \setminus D$, then*

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz = W(\gamma, z_0) f(z_0), \quad z_0 \in D \setminus \Gamma.$$

The proof is simple. We apply the preceding version of the Cauchy integral theorem to the analytic function $g(z) = (f(z) - f(z_0))/(z - z_0)$. From $\int_{\gamma} g(z) dz = 0$, we obtain

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0} dz = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z_0)}{z - z_0} dz = W(\gamma, z_0) f(z_0).$$

Winding numbers can be defined for a finite collection of closed curves, by simply adding the winding numbers for the various curves in the collection. (See the exercises for Section 8.) Thus for instance, if D is a bounded domain with piecewise smooth boundary, we define the winding number $W(\partial D, \zeta)$ of ∂D around $\zeta \notin \partial D$ to be the sum of the winding numbers of the various component curves of ∂D , appropriately oriented. From Cauchy's theorem and the Cauchy integral representation formula,

we then have

$$W(\partial D, \zeta) = \frac{1}{2\pi i} \int_{\partial D} \frac{dz}{z - \zeta} = \begin{cases} 1, & \zeta \in D, \\ 0, & \zeta \notin D \cup \partial D. \end{cases}$$

Thus the winding number of ∂D around points ζ outside of $D \cup \partial D$ is 0, and the winding number jumps to +1 as ζ crosses over ∂D into D . This is a special case of the jump theorem for Cauchy integrals treated in the next section.

Exercises for VIII.6

1. Sketch the closed path $\gamma(t) = e^{it} \sin(2t)$, $0 \leq t \leq 2\pi$, and determine the winding number $W(\gamma, \zeta)$ for each point ζ not on the path.
2. Sketch the closed path $\gamma(t) = e^{-2it} \cos t$, $0 \leq t \leq 2\pi$, and determine the winding number $W(\gamma, \zeta)$ for each point ζ not on the path.
3. Let $f(z)$ be analytic on an open set containing a closed path γ , and suppose $f(z) \neq 0$ on γ . Show that the increase in $\arg f(z)$ around γ is $2\pi W(f \circ \gamma, 0)$.
4. Let D be a domain, and suppose z_0 and z_1 lie in the same connected component of $\mathbb{C} \setminus D$. (a) Show that the increase in the argument of $f(z) = (z - z_0)(z - z_1)$ around any closed curve in D is an even multiple of 2π . (b) Show that $(z - z_0)(z - z_1)$ has an analytic square root in D . (c) Show by example that $(z - z_0)(z - z_1)$ does not necessarily have an analytic cube root in D .
5. Show that if γ is a closed curve in the complex plane, with trace Γ , and if $z_0 \notin \Gamma$, then

$$\int_{\gamma} \frac{1}{(z - z_0)^n} dz = 0, \quad n \geq 2.$$

6. Let γ be a closed path in a domain D such that $W(\gamma, \zeta) = 0$ for all $\zeta \notin D$. Suppose that $f(z)$ is analytic on D except possibly at a finite number of isolated singularities $z_1, \dots, z_m \in D \setminus \Gamma$. Show that

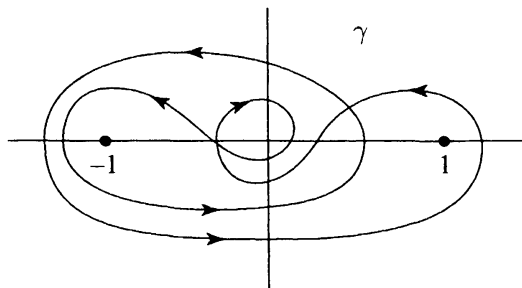
$$\int_{\gamma} f(z) dz = 2\pi i \sum W(\gamma, z_k) \operatorname{Res}[f, z_k].$$

Hint. Consider the Laurent decomposition at each z_k , and use Exercise 5.

7. Evaluate

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z(z^2 - 1)},$$

where γ is the closed path indicated in the figure. *Hint.* Either use Exercise 6, or proceed directly with partial fractions.



8. Let $\gamma(t)$ and $\sigma(t)$, $a \leq t \leq b$, be closed paths. (a) Show that if $\zeta \in \mathbb{C}$ does not lie on the straight line segment between $\gamma(t)$ and $\sigma(t)$, for $a \leq t \leq b$, then $W(\sigma, \zeta) = W(\gamma, \zeta)$. (b) Show that if $|\sigma(t) - \gamma(t)| < |\zeta - \gamma(t)|$ for $a \leq t \leq b$, then $W(\sigma, \zeta) = W(\gamma, \zeta)$.
9. Let $f(z)$ be a continuous complex-valued function on the complex plane such that $f(z)$ is analytic for $|z| < 1$, $f(z) \neq 0$ for $|z| \geq 1$, and $f(z) \rightarrow 1$ as $z \rightarrow \infty$. Show that $f(z) \neq 0$ for $|z| < 1$.
10. Let K be a nonempty closed bounded subset of the complex plane, and let $f(z)$ be a continuous complex-valued function on the complex plane that is analytic on $\mathbb{C} \setminus K$ and at ∞ . Show that every value attained by $f(z)$ on $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$ is attained by $f(z)$ somewhere on K , that is, $f(\mathbb{C}^*) = f(K)$.

7. The Jump Theorem for Cauchy Integrals

Suppose γ is a piecewise smooth curve with trace Γ , and let $f(z)$ be a continuous function on Γ . We define the **Cauchy integral of $f(z)$ along γ** to be the function $F(\zeta)$ defined off Γ by

$$(7.1) \quad F(\zeta) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - \zeta} dz, \quad \zeta \in \mathbb{C} \setminus \Gamma.$$

We have seen that the Cauchy integral $F(\zeta)$ is analytic on $\mathbb{C} \setminus \Gamma$, and $F(\zeta)$ vanishes at ∞ .

Example. If γ is a closed curve parameterizing the (positively oriented) boundary Γ of a domain D , and if $f(z)$ is analytic on D and across Γ , then the Cauchy integral $F(\zeta)$ of $f(z)$ around γ is given by

$$F(\zeta) = \begin{cases} f(\zeta), & \zeta \in D, \\ 0, & \zeta \in \mathbb{C} \setminus (D \cup \Gamma). \end{cases}$$

This follows from Cauchy's theorem if $\zeta \in \mathbb{C} \setminus (D \cup \Gamma)$, and from the Cauchy integral formula if $\zeta \in D$.

Example. If γ is a closed path, the Cauchy integral of the constant function $f(z) \equiv 1$ around γ is the winding number $F(\zeta) = W(\gamma, \zeta)$ of γ around ζ .

Now suppose that the continuous function $f(z)$ on Γ is analytic at some point $z_0 \in \Gamma$. We suppose that there is a small disk U centered at z_0 such that $f(z)$ is analytic on $U \cup \partial U$ and such that γ passes through U in a piecewise smooth subarc γ_0 traveling from $z_1 \in \partial U$ to another point $z_2 \in \partial U$, so that $\gamma \setminus \gamma_0$ does not enter U , while the trace Γ_0 of γ_0 divides U into two connected open domains. As we traverse γ_0 in the positive direction, one of these domains U_+ lies on the right side of γ_0 , the other U_- on the left side. We break the integral for $F(\zeta)$ into two pieces

$$F(\zeta) = \frac{1}{2\pi i} \int_{\gamma_0} \frac{f(z)}{z - \zeta} dz + \frac{1}{2\pi i} \int_{\gamma \setminus \gamma_0} \frac{f(z)}{z - \zeta} dz = F_0(\zeta) + G(\zeta).$$

The second integral $G(\zeta)$ is analytic on U , while the first integral $F_0(\zeta)$ is analytic on $U \setminus \Gamma_0$. Let γ_+ denote the arc of ∂U from z_1 to z_2 that lies in the boundary of U_+ , and let γ_- denote the arc of ∂U from z_1 to z_2 that lies in the boundary of U_- , as in the next figure. We define

$$F_+(\zeta) = \frac{1}{2\pi i} \int_{\gamma_-} \frac{f(z)}{z - \zeta} dz + G(\zeta), \quad \zeta \in U.$$

By Cauchy's theorem,

$$\frac{1}{2\pi i} \int_{\gamma_0} \frac{f(z)}{z - \zeta} dz = \frac{1}{2\pi i} \int_{\gamma_-} \frac{f(z)}{z - \zeta} dz, \quad \zeta \in U_+,$$

and consequently, $F(\zeta) = F_+(\zeta)$ for $\zeta \in U_+$. Similarly, we define

$$F_-(\zeta) = \frac{1}{2\pi i} \int_{\gamma_+} \frac{f(z)}{z - \zeta} dz + G(\zeta), \quad \zeta \in U,$$

and then from Cauchy's theorem we obtain $F(\zeta) = F_-(\zeta)$ for $\zeta \in U_-$. If we integrate along the arc γ_+ of ∂U from z_1 to z_2 , then backwards along the arc γ_- of ∂U from z_2 to z_1 , we obtain the integral around the boundary ∂U . This and Cauchy's theorem yield

$$F_-(\zeta) - F_+(\zeta) = \frac{1}{2\pi i} \int_{\partial U} \frac{f(z)}{z - \zeta} dz = f(\zeta), \quad \zeta \in U.$$

We have proved the following.

Theorem (Jump Theorem for Cauchy Integrals). Suppose that Γ is a smooth curve that passes through z_0 , and $f(z)$ is a continuous function on Γ that is analytic at z_0 . Let U be a small disk containing z_0 such that $f(z)$ is analytic on U and such that Γ divides U into the two components U_+ and U_- as above. Then there are analytic functions $F_+(\zeta)$ and $F_-(\zeta)$ on U satisfying

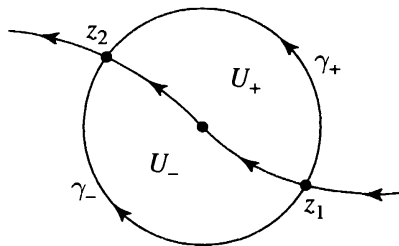
$$F_-(\zeta) - F_+(\zeta) = f(\zeta), \quad \zeta \in U,$$

and such that the Cauchy integral $F(\zeta)$ of $f(z)$, defined by (7.1), satisfies

$$F(\zeta) = \begin{cases} F_+(\zeta), & \zeta \in U_+, \\ F_-(\zeta), & \zeta \in U_-. \end{cases}$$

Thus as ζ crosses Γ from right to left (from U_+ to U_-), the values of the Cauchy integral $F(\zeta)$ jump by $f(\zeta)$,

$$F_-(\zeta) = F_+(\zeta) + f(\zeta), \quad \zeta \in U.$$



Example. In the first example above, where γ parameterizes the boundary Γ of a domain D and $f(z)$ is analytic on D and across Γ , the Cauchy integral of $f(z)$ jumps from 0 outside Γ to $f(\zeta)$ inside Γ .

Now consider the special case where $f(z) \equiv 1$ and γ is a closed curve. In this case, the Cauchy integral (7.1) is the winding number $W(\gamma, \zeta)$, and we are led to the following jump theorem.

Theorem (Jump Theorem for Winding Numbers). *Let γ be a closed path with trace Γ , and let $z_0 = \gamma(t_0) \in \Gamma$. Suppose that $\gamma(t)$ is continuously differentiable for t near t_0 , with $\gamma'(t_0) \neq 0$. Let U be a small disk centered at z_0 that is divided into two components $U_{\pm}$ by a segment γ_0 on γ as above, and suppose that $\gamma \setminus \gamma_0$ does not enter U . Then as $\zeta \in U$ crosses Γ from right to left (from U_+ to U_-), the winding number $W(\gamma, \zeta)$ jumps by +1,*

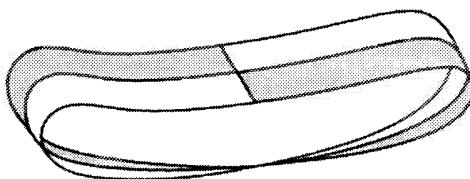
$$W(\gamma, \zeta_-) = W(\gamma, \zeta_+) + 1, \quad \zeta_- \in U_-, \zeta_+ \in U_+.$$

This does not follow immediately from the preceding jump theorem, since the path $\gamma \setminus \gamma_0$ is not necessarily smooth. However, the winding number can be expressed as

$$W(\gamma, \zeta) = \frac{1}{2\pi i} \int_{\gamma_0} \frac{dz}{z - \zeta} + G(\zeta),$$

where $G(\zeta)$ is the increase of the argument of $z - \zeta$ along $\gamma \setminus \gamma_0$. If we apply the jump theorem to the Cauchy integral along γ_0 , and we observe that $G(\zeta)$ is analytic on U and hence does not jump, we obtain the jump theorem for the winding number.

Recall (Section III.1) that a closed path Γ in the complex plane is **simple** if it is parameterized by a continuous function $\gamma(t)$, $a \leq t \leq b$, such that $\gamma(s) \neq \gamma(t)$ for $a \leq s < t < b$. We refer to a simple closed path also as a **simple closed curve**. By regarding the interval $[a, b]$ with its endpoints identified as a circle, we can think of a simple closed curve as a continuous one-to-one image of a circle. The Jordan curve theorem asserts that a simple closed curve in the complex plane divides the plane into exactly two connected components, one bounded (the “inside”) and the other unbounded (the “outside”). While the theorem may seem intuitively plausible, a glance at the figure for Exercise 6 should convince one that the theorem is not completely obvious. In fact, Jordan did not give a rigorous proof of the Jordan curve theorem, and the known proofs of the theorem require a fair amount of effort.



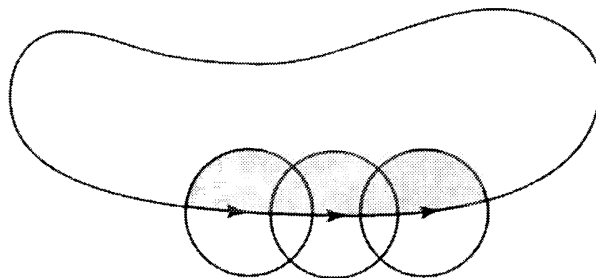
Möbius band

To appreciate the sort of difficulty that might arise, consider the Möbius band, obtained from a long thin strip of paper by flipping over one end and then gluing the two ends together. If one cuts the strip along the median line with a scissors, the resulting surface is connected. Thus the median line, which is a simple closed curve on the surface, does not divide the surface into two connected components. The median line has only one “side.”

The proof of the Jordan curve theorem for piecewise smooth curves is substantially easier than the proof for arbitrary simple closed curves. We give a proof based on the jump theorem for the winding number. This idea forms the basis for a proof in the general case, which is laid out in the exercises.

Suppose γ is a simple closed curve such that $\gamma(t)$ is continuously differentiable on some parameter interval. Then we can find z_0 on γ and a disk U centered at z_0 such that the smooth segment divides U into two pieces as above. (This requires a nonzero tangent vector and some elementary analysis at the level of the implicit function theorem.) Since the winding number $W(\gamma, \zeta)$ jumps as ζ crosses the segment of γ in U , it attains at least two values on $\mathbb{C} \setminus \Gamma$. Since it is constant on each component of $\mathbb{C} \setminus \Gamma$, the complement of Γ has at least two connected components.

If the entire curve γ is smooth, or even just piecewise smooth, we can cover Γ by a finite number of open disks U as above, each of which is divided into two pieces by γ . By tracking successive components of $U \setminus \Gamma$ corresponding to consecutive U 's, we see that each point in any $U \setminus \Gamma$ can be



joined by a path in the $U \setminus \Gamma$'s to one or the other component of any other $U \setminus \Gamma$. It follows that each point of $\mathbb{C} \setminus \Gamma$ can be joined by a path in $\mathbb{C} \setminus \Gamma$ to one or the other side of some fixed U , and consequently $\mathbb{C} \setminus \Gamma$ has at most two components. We have proved the following.

Theorem (Jordan Curve Theorem for Smooth Curves). *Let γ be a piecewise smooth simple closed curve in the complex plane $\mathbb{C}$, and let $\Gamma = \gamma([a, b])$ be its trace. Then Γ divides the complex plane into two connected components, one bounded and the other unbounded, each of which has Γ as its boundary. Further, $W(\gamma, \zeta) = 0$ if ζ is in the unbounded component of $\mathbb{C} \setminus \Gamma$, and $W(\gamma, \zeta) = \pm 1$ if ζ is in the bounded component of $\mathbb{C} \setminus \Gamma$, where the choice of sign depends on the orientation of γ .*

Exercises for VIII.7

1. Let $f(z)$ be an entire function, and suppose $g(\zeta)$ is analytic for ζ in the open upper and lower half-planes and across the interval $(-1, 1)$ on the real line. Suppose that

$$\int_{-1}^1 \frac{f(x)}{x - \zeta} dx = g(\zeta)$$

for ζ in the upper half-plane. What is the value of the integral when ζ is in the lower half-plane? Justify your answer carefully.

2. Show that

$$\int_{-1}^1 \frac{dx}{x - \zeta} = \operatorname{Log} \left(\frac{\zeta - 1}{\zeta + 1} \right), \quad \zeta \in \mathbb{C} \setminus [-1, +1].$$

(Note that we use the principal branch of the logarithm here.) Reconcile this result with your solution to Exercise 1.

3. Find the Cauchy integrals of the following functions around the unit circle $\Gamma = \{|z| = 1\}$, positively oriented. (a) z , (b) $\frac{1}{z}$, (c) $x = \operatorname{Re}(z)$, (d) $y = \operatorname{Im}(z)$.
4. Suppose $f(z)$ is analytic on an annulus $\{\rho < |z| < \sigma\}$, and let $f(z) = f_0(z) + f_1(z)$ be the Laurent decomposition of $f(z)$. (See Section VI.1.) Fix r between ρ and σ , and let $F(\zeta)$ be the Cauchy integral of $f(z)$ around the circle $|z| = r$. Show that $f_0(\zeta) = F(\zeta)$

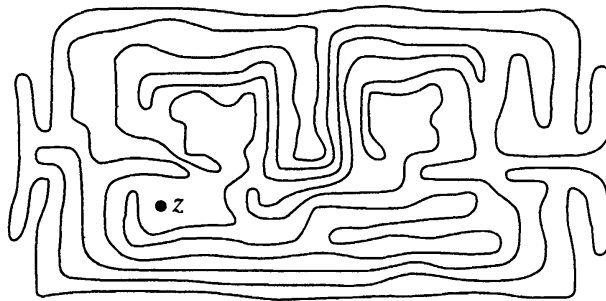
for $|\zeta| < r$, and $f_1(\zeta) = -F(\zeta)$ for $|\zeta| > r$. Show further that $f_0(\zeta) = F_-(\zeta)$ and $f_1(\zeta) = -F_+(\zeta)$. *Remark.* The formula $f(z) = f_0(z) + f_1(z)$ reflects the jump theorem for the Cauchy integral of $f(z)$ around circles $|z| = r$.

5. Let γ be a piecewise smooth curve, and let $F(\zeta)$ be the Cauchy integral (7.1) of a continuous function $f(z)$ on γ . Show that if $g(z)$ is a smooth function on the complex plane that is zero off some bounded set, then

$$\int_{\gamma} g(z)f(z) dz = 2i \iint_{\mathbb{C}} \frac{\partial g}{\partial \bar{z}} F(z) dx dy.$$

Hint. Recall Pompeiu's formula (Section IV.8).

6. Determine whether the point z lies inside or outside. Explain.



7. A **simple arc** Γ in $\mathbb{C}$ is the image of a continuous one-to-one function $\gamma(t)$ from a closed interval $[a, b]$ to the complex plane. Show that a simple arc Γ in $\mathbb{C}$ has a connected complement, that is, $\mathbb{C} \setminus \Gamma$ is connected. You may use the Tietze extension theorem, that a continuous real-valued function on a closed subset of the complex plane can be extended to a continuous real-valued function on the entire complex plane. *Hint.* Suppose z_0 belongs to a bounded component of $\mathbb{C} \setminus \Gamma$. Find a continuous determination $h(z)$ of $\log(z - z_0)$ on Γ , extend $h(z)$ to a continuous function on $\mathbb{C}^*$, and define $f(z) = z - z_0$ on the component of $\mathbb{C} \setminus \Gamma$ containing z_0 , and $f(z) = e^{h(z)}$ on the remainder of $\mathbb{C}^* \setminus \Gamma$. Consider the increase in the argument of $f(z)$ around circles centered at z_0 .
8. Prove the Jordan curve theorem for a simple closed curve γ by filling in the following proof outline.
- Show that each component of $\mathbb{C} \setminus \Gamma$ has boundary Γ . *Hint.* For $z_0 = \gamma(t_0) \in \Gamma$, apply the preceding exercise to the simple arc $\Gamma \setminus \gamma(I)$, where I is a small open parameter interval containing t_0 .
 - Prove the Jordan curve theorem in the case where γ contains a straight line segment.

- (c) Show that for any $z_0 = \gamma(t_0) \in \Gamma$, any small disk D_0 containing z_0 , and any component U of $\mathbb{C} \setminus \Gamma$, there are points $z_1 = \gamma(t_1)$ and $z_2 = \gamma(t_2)$ such that the image of the parameter segment between t_1 and t_2 is contained in D_0 and such that z_1 and z_2 can be joined by a broken line segment in $U \cap D_0$.
- (d) With notation as in (b), let σ be the simple closed curve obtained by replacing the segment of γ in D_0 between z_0 and z_1 by the broken line segment in $U \cap D_0$ between them, and let τ be the simple closed curve in D_0 obtained by following the segment of γ in D_0 from z_0 to z_1 and returning to z_0 along the broken line segment. Show that $W(\tau, \zeta) = 0$ and $W(\gamma, \zeta) = W(\sigma, \zeta)$ for $\zeta \in \mathbb{C} \setminus \Gamma$, $\zeta \notin D_0$.
- (e) Using (b) and (d), show that $\mathbb{C} \setminus \Gamma$ has at least two components and that $W(\gamma, \zeta) = \pm 1$ for ζ in each bounded component of $\mathbb{C} \setminus \Gamma$.
- (f) By taking U in (c) to be a bounded component of $\mathbb{C} \setminus \Gamma$, show that $W(\gamma, \zeta) = 0$ for ζ in any other component of $\mathbb{C} \setminus \Gamma$.

8. Simply Connected Domains

A domain in the plane is “simply connected” if it has no “holes.” Disks and rectangles are simply connected. More generally, star-shaped domains are simply connected. Annuli, punctured disks, and the punctured plane have “holes” and are *not* simply connected. Our aim in this section is to make more precise the notion of “simple connectedness.”

Let $\gamma(t)$, $a \leq t \leq b$, be a closed path in a domain D . We say that γ is **deformable to a point** if there are paths $\gamma_s(t)$, $a \leq t \leq b$, $0 \leq s \leq 1$, in D such that $\gamma_s(t)$ depends continuously on both s and t , $\gamma_0 = \gamma$, and $\gamma_1(t) \equiv z_1$ is the constant path at some point $z_1 \in D$. The domain D is **simply connected** if every closed path in D can be deformed to a point.

Example. If D is a star-shaped domain with respect to z_0 , then any closed path γ in D can be continuously deformed to the point z_0 by pulling the path to z_0 along straight line segments. We set

$$\gamma_s(t) = sz_0 + (1-s)\gamma(t), \quad a \leq t \leq b, \quad 0 \leq s \leq 1,$$

and then the closed paths γ_s deform $\gamma_0 = \gamma$ to the point path $\gamma_1 \equiv z_0$. Thus D is simply connected.

It turns out that if a closed path can be deformed to a point, then it can be deformed to a point in such a way that each path in the deformation starts and ends at the same fixed point. That is the content of the following lemma.

Lemma. Let $\gamma(t)$, $0 \leq t \leq 1$, be a closed path in D , with $z_0 = \gamma(0) = \gamma(1)$. Suppose that γ can be deformed continuously to a point in D . Then there is a continuous family of closed paths γ_s , $0 \leq s \leq 1$, such that $\gamma_0 = \gamma$, γ_1 is the constant path at z_0 , and each path γ_s starts and ends at z_0 .

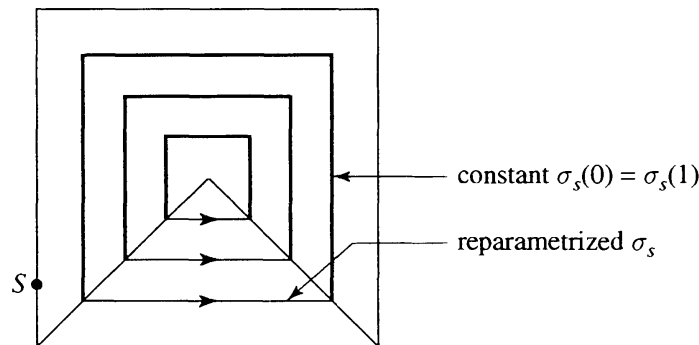
Suppose that γ is deformed to a point by closed paths σ_s , $0 \leq s \leq \frac{1}{2}$. Thus $\sigma_0(t) = \gamma(t)$ for $0 \leq t \leq 1$, $\sigma_{1/2}(t) = z_1$ for $0 \leq t \leq 1$, and $\sigma_s(0) = \sigma_s(1)$ for $0 \leq s \leq \frac{1}{2}$. We deform γ to the point z_0 by means of paths γ_s , $0 \leq s \leq 1$, that all start and terminate at z_0 , in two stages. In the first stage we deform γ to a path $\gamma_{1/2}$ that follows the starting points $\sigma_s(0)$ of σ_s from z_0 to z_1 , and then reverses direction and follows them backward from z_1 to z_0 . Such a deformation γ_s is given explicitly for $0 \leq s < \frac{1}{2}$ by

$$\gamma_s(t) = \begin{cases} \sigma_t(0), & 0 \leq t \leq s, \\ \sigma_s((t-s)/(1-2s)), & s \leq t \leq 1-s, \\ \sigma_{1-t}(1), & 1-s \leq t \leq 1. \end{cases}$$

For the second stage we deform $\gamma_{1/2}$ to the point path at z_0 by running partway along $\gamma_{1/2}$, pausing, then returning, with increasing pauses. Such a deformation γ_s , $\frac{1}{2} \leq s \leq 1$, is given explicitly by

$$\gamma_s(t) = \begin{cases} \sigma_t(0), & 0 \leq t \leq 1-s, \\ \sigma_{1-s}(0), & 1-s \leq t \leq s, \\ \sigma_{1-t}(0), & s \leq t \leq 1. \end{cases}$$

The definition of the function $\gamma_s(t)$, regarded as a function on the square, is suggested in the figure. On the horizontal intervals in the bottom half, $\gamma_s(t)$ is a reparametrization of $\sigma_s(t)$, while on each horizontal interval in the top half and on each vertical interval $\gamma_s(t)$ is constant.



Another topological notion we require is connectedness. We have already defined what it means for an open subset of the complex plane to be connected, namely, that any two points can be joined by a path. In the context of general topological spaces this concept is defined to be “path connectedness.” For arbitrary topological spaces there is another definition of connectedness, which we will use in this section. It is equivalent to path connectedness for open sets in the plane.

Let X be a subset of the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$. We say that X is **connected** if every continuous integer-valued function on X is constant. Thus X is *not* connected if and only if there is a subset E of X such that both E and $X \setminus E$ are nonempty, and no sequence in E has a limit in $X \setminus E$, nor does any sequence in $X \setminus E$ have a limit in E . If there is such a set E , then the function that is 1 on E and 0 on $X \setminus E$ is a continuous integer-valued function on X that is not constant. Conversely, if there is a nonconstant continuous integer-valued function on X , we take E to be one of the level sets of the function, and we see that X is not connected. Note that if X is a closed subset of $\mathbb{C}^*$ that is not connected, we can take E above such that $\infty \notin E$, and then E is a compact subset of the complex plane with positive distance from $X \setminus E$, that is, there is $\delta > 0$ such that $|z - w| \geq \delta$ for all $z \in E$ and $w \in X \setminus E$.



For domains in the plane many conditions can be formulated that are equivalent to simple connectivity. We give several that are useful for our purposes.

Theorem. *The following properties are equivalent, for a domain D in the complex plane:*

- (i) D is simply connected,
- (ii) every closed differential on D is exact,
- (iii) for each $z_0 \in \mathbb{C} \setminus D$, there is an analytic branch of $\log(z - z_0)$ defined on D ,
- (iv) each closed curve γ in D has winding number $W(\gamma, z_0) = 0$ about all points $z_0 \in \mathbb{C} \setminus D$,
- (v) the complement of D in the extended complex plane $\mathbb{C}^*$ is connected.

In practice, the condition (v) is usually the easiest to check for a given domain D . It is usually easy to spot any set E in $\mathbb{C} \setminus D$ that has positive distance from the rest of $\mathbb{C} \setminus D$. We think of such sets E as representing holes in D .

Example. If D is an annulus $\{r < |z| < s\}$, we can take E to be the closed disk $\{|z| \leq r\}$. Then the rest of $\mathbb{C} \setminus D$ is the exterior set $\{|z| \geq s\}$, every point of which has distance at least $s - r$ from E . Consequently, $\mathbb{C} \setminus D$ is not connected, and D is not simply connected. The set $\{|z| \leq r\}$ is a hole in D . Similarly, if D is the punctured disk $\{0 < |z| < 1\}$ or the punctured plane $\mathbb{C} \setminus \{0\}$, we can take E to be the puncture $\{0\}$, and we see that D is not simply connected.

Now we turn to the proof of the theorem. The implications that are trivial or easy consequences of our previous discussions are (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Leftrightarrow$ (iv) and (v) $\Rightarrow$ (iv). New ideas are required to establish the implications (iv) $\Rightarrow$ (v) and (iv) $\Rightarrow$ (i).

Proof that (i) $\Rightarrow$ (ii): This follows from our discussion of deformation of paths in Section III.2. Let $Pdx + Qdy$ be a closed differential on D . According to the discussion in Section III.2, $\int_{\gamma} Pdx + Qdy = \int_{\sigma} Pdx + Qdy$ for any closed paths γ and σ in D such that γ can be deformed continuously to σ . Since the integral over a one-point path is zero, the simple connectivity of D implies that $\int_{\gamma} Pdx + Qdy = 0$ for any closed path in D . Consequently, $Pdx + Qdy$ is independent of path, hence exact.

Proof that (ii) $\Rightarrow$ (iii): If every closed differential on D is exact, then in particular for fixed $z_0 \in \mathbb{C} \setminus D$ the differential $d \arg(z - z_0)$ is exact. We write $d \arg(z - z_0) = dh(z)$ for some smooth function $h(z)$, normalized so that $h(z_1)$ is a value of $\arg(z_1 - z_0)$ for some fixed point $z_1 \in D$. Then $h(z)$ is a continuous determination of $\arg(z - z_0)$ in D , and $\log|z - z_0| + ih(z)$ is an analytic branch of $\log(z - z_0)$ in D .

Proof that (iii) $\Leftrightarrow$ (iv): If there is an analytic branch $f(z)$ of $\log(z - z_0)$, then

$$W(\gamma, z_0) = \frac{1}{2\pi i} \int_{\gamma} df = 0$$

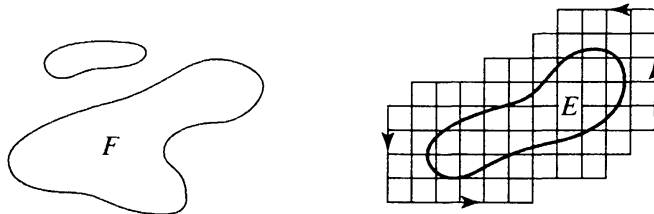
for any closed path γ in D . Conversely, if $W(\gamma, z_0) = 0$ for any closed path γ in D , then the analytic differential $dz/(z - z_0)$ is independent of path, and there is an analytic function $f(z)$ such that $df = dz/(z - z_0)$. An analytic branch of $\log(z - z_0)$ is obtained by adding an appropriate constant to $f(z)$.

Proof that (v) $\Rightarrow$ (iv): Let γ be a closed path in D . The winding number $W(\gamma, \zeta)$ is a continuous integer-valued function on $\mathbb{C} \setminus D$ that is zero for large ζ . If we set it equal to 0 for $\zeta = \infty$, we obtain a continuous integer-valued function on $\mathbb{C}^* \setminus D$ that vanishes at ∞ . Since $\mathbb{C}^* \setminus D$ is connected, the function is identically zero. Hence $W(\gamma, \zeta) = 0$ for $\zeta \in \mathbb{C} \setminus D$.

Proof that (iv) $\Rightarrow$ (v): If $\mathbb{C}^* \setminus D$ is not connected, then there are a closed bounded subset E of $\mathbb{C} \setminus D$ and $\delta > 0$ such that $|z - w| \geq 4\delta$ for every point $z \in E$ and every point $w \in F = \mathbb{C} \setminus (D \cup E)$. Fix a point $z_0 \in E$, and cover the complex plane by a grid of squares of side length δ so that z_0 lies in the interior of one of the squares. Let K be the union of all the (closed) squares in the grid that contain a point of distance at most δ from E . Then K is a finite union of closed squares, K is disjoint from F , and the boundary ∂K of K does not contain points of either E or F . Thus ∂K is a finite union of straight line segments contained in D . Further, if we denote by K_j the squares from the grid in K , then

$$\int_{\partial K} d \arg(z - z_0) = \sum \int_{\partial K_j} d \arg(z - z_0) = 2\pi,$$

since the only summand that is not zero is the integral over the boundary of the square containing z_0 . Now, ∂K is a union of a finite number of closed paths in D . Thus there is a closed path γ in D obtained by following consecutively sides of squares in the grid, for which $\int_{\gamma} d \arg(z - z_0) \neq 0$. Consequently, $W(\gamma, z_0) \neq 0$.

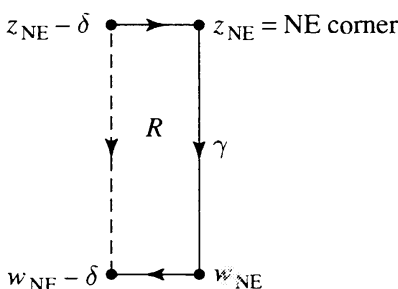


Proof that (iv) $\implies$ (i): Our proof is based on the “northeast corner” argument. We can deform any closed γ in D to a path that follows the sides of a grid of squares of small side length $\delta > 0$. Thus we fix δ , and we consider closed paths in D that are obtained by following consecutively sides of squares in the grid, so that the path can be represented as a sequence of sides, each with a direction. We proceed by induction on the number of consecutive sides in the path. Thus we assume that γ is represented by a sequence of sides of the grid, and that any closed path formed by a shorter sequence of sides in the grid can be deformed in D to a point. We must show that γ can be deformed in D to a point.

Suppose first that the closed path γ intersects itself, so that γ is obtained by traversing successively two shorter closed paths, each of which starts and terminates at a vertex z_0 of the grid. By our induction hypothesis, each of the shorter paths can be deformed in D to a point. By the lemma, each of the paths can be deformed in D to the point z_0 by means of closed paths that begin and end at z_0 . By traversing successively the paths in the two deformations, we can then deform the path γ in D to the point z_0 .

We may suppose, then, that the closed path γ does not intersect itself. In the path γ consider the vertices furthest to the right, for which $\operatorname{Re}(z)$ is the largest, and among these let z_{NE} be the highest vertex, for which $\operatorname{Im}(z)$ is the largest. Reversing the direction of the path if necessary, we can assume that the path follows the horizontal edge from $z_{\text{NE}} - \delta$ to z_{NE} , where it makes a right turn and proceeds down to $w_{\text{NE}} = z_{\text{NE}} - i\delta$, then makes another right turn and proceeds to $w_{\text{NE}} - \delta$, as in the figure. Let R be the open rectangle with vertices $z_{\text{NE}} - \delta$, z_{NE} , w_{NE} , and $w_{\text{NE}} - \delta$. Since $W(\gamma, \zeta) = 0$ for ζ in the unbounded component of the complement of γ , we have $W(\gamma, \zeta) = 0$ just to the right of the segment from z_{NE} to w_{NE} . By the jump theorem for the winding number (Section 3), we have $W(\gamma, \zeta) = -1$ for ζ to the left of the vertical segment, hence for all $\zeta \in R$. It follows that R is contained in D . Further, by continuity we have $W(\gamma, \zeta) = -1$ at all points of the left vertical side of R that are not on γ , so that none of the points of the boundary of R belong to $\mathbb{C} \setminus D$, and ∂R is also contained

in D . Let σ be the path obtained from γ by replacing the segments of γ consisting of the three edges of the rectangle R by the vertical edge of R from $z_{NE} - \delta$ to $w_{NE} - \delta$. Then γ can be deformed to σ in D , simply by deforming the three sides of the rectangle to the fourth side. Since σ is shorter than γ , σ can be deformed in D to a point. Thus γ can also be deformed in D to a point. This completes the proof of the implication and of the theorem.



In Chapter XI we will prove the Riemann mapping theorem, which asserts that every simply connected domain in the plane except the plane itself can be mapped conformally onto the open unit disk. What we will actually establish in Section XI.3 is that if $D \neq \mathbb{C}$ and if (iii) holds, then there is a conformal map of D onto the open unit disk. From this it follows easily that a domain D satisfying (iii) is simply connected (see Exercise 7), and also that D has a connected complement. Thus the proof of the Riemann mapping theorem will provide us with an alternative method to show that (iii) implies (i) and (v). The proof given above has the advantage of being “elementary” in nature, whereas the proof of the Riemann mapping theorem depends on the arguably nonelementary notion of compactness of families of analytic functions.

Exercises for VIII.8

- Which of the following domains in $\mathbb{C}$ are simply connected? Justify your answers. (a) $D = \{\operatorname{Im} z > 0\} \setminus [0, i]$, the upper half-plane with a vertical slit from 0 to i . (b) $D = \{\operatorname{Im} z > 0\} \setminus [i, 2i]$, the upper half-plane with a vertical slit from i to $2i$. (c) $D = \mathbb{C} \setminus [0, +\infty)$, the complex plane slit along the positive real axis. (d) $D = \mathbb{C} \setminus [-1, 1]$, the complex plane with an interval deleted.
- Show that a domain D in the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$ is simply connected if and only if its complement $\mathbb{C}^* \setminus D$ is connected. *Hint.* If $D \neq \mathbb{C}^*$, move a point in the complement of D to ∞ . If $D = \mathbb{C}^*$, first deform a given closed path to one that does not cover the sphere, then deform it to a point by pulling along arcs of great circles.
- Which of the following domains in $\mathbb{C}^*$ are simply connected? Justify your answers. (a) $D = \mathbb{C}^* \setminus [-1, 1]$, the extended complex plane with

an interval deleted, (b) $D = \mathbb{C}^* \setminus \{-1, 0, 1\}$, the thrice-punctured sphere.

4. Show that a domain D in the complex plane is simply connected if and only if any analytic function $f(z)$ on D that does not vanish at any point of D has an analytic logarithm on D . *Hint.* If $f(z) \neq 0$ on D , consider the function

$$G(z) = \int_{z_0}^z \frac{f'(w)}{f(w)} dw.$$

5. Show that a domain D is simply connected if and only if any analytic function $f(z)$ on D that does not vanish at any point of D has an analytic square root on D . Show that this occurs if and only if for any point $z_0 \notin D$ the function $z - z_0$ has an analytic square root on D .
6. Show that a domain D is simply connected if and only if each continuous function $f(z)$ on D that does not vanish at any point of D has a continuous logarithm on D .
7. Let E be a closed connected subset of the extended complex plane $\mathbb{C}^*$. Show that each connected component of $\mathbb{C}^* \setminus E$ is simply connected.
8. Show that simple connectivity is a “topological property,” that is, if U and V are domains, and φ is a continuous map of U onto V such that φ^{-1} is also continuous, then U is simply connected if and only if V is simply connected.
9. Suppose that $f(z)$ is analytic on a domain D , and $f'(z)$ has no zeros on D . Suppose also that $f(D)$ is simply connected, and that there is a branch $g(w)$ of f^{-1} that is analytic at $w_0 = f(z_0)$ and that can be continued analytically along any path in $f(D)$ starting at w_0 . Show that $f(z)$ is one-to-one on D .
10. We define an **integral 1-cycle** in D to be an expression of the form $\sigma = \sum k_j \gamma_j$, where $\gamma_1, \dots, \gamma_m$ are closed paths in D and $k_1, \dots, k_m$ are integers. We define the **winding number of σ about ζ** to be $W(\sigma, \zeta) = \sum k_j W(\gamma_j, \zeta)$, $\zeta \in \mathbb{C} \setminus D$. Show that if $h(\zeta)$ is a continuous integer-valued function on $\mathbb{C}^* \setminus D$ such that $h(\infty) = 0$, then there is an integral 1-cycle σ on D such that $W(\sigma, \zeta) = h(\zeta)$ for all $\zeta \in \mathbb{C} \setminus D$.
11. An integral 1-cycle σ is **homologous to zero** in D if $W(\sigma, \zeta) = 0$ whenever $\zeta \notin D$. Let U be a bounded domain whose boundary consists of a finite number of piecewise smooth closed curves $\gamma_1, \dots, \gamma_m$, oriented positively with respect to U , such that U together with its